Telematics and Informatics Reports 22 (2026) 100325

Contents lists available at ScienceDirect

Telematics and Informatics Reports

journal homepage: www.elsevier.com/locate/teler

Healthcare digital twins: A methodological literature review on integrating iot and AI for personalized medicine and predictive care

Sara Shahnazinia [a], Mahsa Tavasoli [a], Abdolhossien Sarrafzadeh [b,*], Ali Karimoddini [c]

[a] Department of Applied Science and Technology, North Carolina Agricultural and Technical State University, Greensboro, NC 27411, USA
[b] Department of Electrical and Computer Engineering, Old Dominion University, Norfolk, VA 23529, USA
[c] Department of Electrical and Computer Engineering, North Carolina Agricultural and Technical State University, Greensboro, NC 27411, USA

ARTICLE INFO

Keywords:
Digital Twin (DT)
IoT-AI healthcare
Predictive healthcare
Healthcare data integration
Medical simulation
Ethical-regulatory frameworks

ABSTRACT

Digital Twin (DT) technology has the potential to revolutionize healthcare delivery and enhance patient outcomes through personalized and precision medicine, simulation models for operations and interventions, and drug discovery. However, successful implementation of DTs in Internet of Things (IoT) and artificial intelligence (AI) healthcare is contingent upon addressing key challenges such as privacy, ethics, and robust data security. This paper presents a methodological literature review of DT applications in healthcare, systematically analyzing the current state of research, key enabling technologies, and implementation challenges. The review summarizes DT categorization approaches (application-based, technology-based, and real-time function-based); delineates core DT components such as sensors, data pipelines, AI/ML capabilities, security and governance measures; and surveys data collection and sensing technologies spanning EHRs, wearable/IoMT devices, and medical imaging. It further synthesizes diverse case studies across hospital management, diagnosis and treatment, patient monitoring and management, personalized therapies, and medical devices, highlighting both performance gains and translational gaps. Based on the corpus, the review identifies data integration and interoperability across heterogeneous healthcare systems as the foundational barrier to widespread DT adoption; without standardized protocols and semantics for multi-source data fusion and real-time exchange, the promise of adaptive, personalized, and predictive care remains largely unrealized. Finally, we outline actionable directions including standards-aligned data models, privacy-preserving learning (for example, federated or split learning), measurable clinical validation, and workflow-aware user experience design to accelerate translation from prototypes to routine clinical practice.

1. Introduction

Technological developments have continuously shaped various sectors, including healthcare, where the introduction of the Internet of Things (IoT) marks a significant milestone in this evolution [1]. IoT's integration into daily life has revolutionized how we interact with technology, particularly in healthcare, through smartphones, smart homes, and wearable health monitors. IoT sensors and devices have been instrumental in enhancing healthcare systems by facilitating health workflows, speeding up access to medical records, increasing data accuracy, and enabling efficient data sharing [2]. The importance of these advancements is underscored by reports from the U.S. Institute of Medicine, highlighting medical errors as a significant concern, often attributed to data inefficiencies, such as the inability to access a patient's medical history, delayed diagnoses or corrupted health data [3]. IoT has also laid the foundation for more advanced healthcare solutions such as remote patient monitoring, predictive diagnostics, and smart hospital management systems, which allow healthcare providers to make data-driven decisions in real-time [4]. IoT's role in healthcare is not just about connecting devices but also about significantly reducing mortality rates by providing real-time health data. This trend is further bolstered by the rise in personal health monitoring devices that actively track vital health parameters, such as electrocardiogram (ECG), blood pressure, heart rate, and sugar levels, thus reducing potential data recording errors. These devices anonymously capture and transfer data to the cloud for analysis, facilitating symptom detection and alerting health personnel when necessary. The convergence of IoT and healthcare has initiated an intelligent, context-aware era, addressing challenges such as information sharing, diagnostic inefficiency, and medication errors, ultimately enhancing the quality of life and healthcare service standards [5,6].

* Corresponding author.
E-mail addresses: shahnazinia@aggies.ncat.edu (S. Shahnazinia), mtavasoli@aggies.ncat.edu (M. Tavasoli), asarrafz@odu.edu (A. Sarrafzadeh), akarimod@ncat.edu (A. Karimoddini).

https://doi.org/10.1016/j.teler.2026.100325
Received 28 May 2025; Received in revised form 19 March 2026; Accepted 26 March 2026
Available online 31 March 2026
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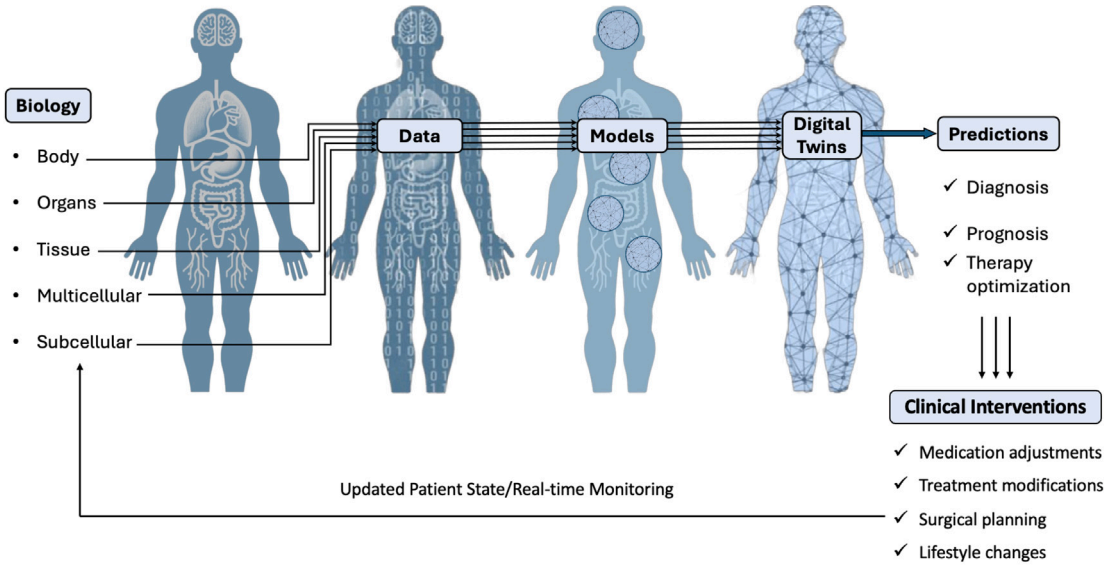


Fig. 1. DT process: from data collection to predictions with continuous feedback loop for treatment optimization and model refinement.

Digital twins technology, identified as one of the top trending technologies by the IEEE Computer Society in 2020, has emerged as a transformative tool in healthcare [7]. Building on the capabilities of IoT, DT technology leverages real-time data from connected devices to create highly detailed digital replicas of physical systems. It involves creating a digital replica of a physical object, such as a patient's body or specific organs, and is powered by a fusion of technologies, including artificial intelligence (AI), data analytics, IoT, and virtual/augmented reality [8,9]. Fuller et al. [10] investigated the concept of DT, which refers to a virtual representation of a physical object or system. Their research focused on the enabling technologies, challenges, and open research questions related to developing and implementing DTs. The study identified three main areas where DTs have the potential to be highly effective: smart cities, manufacturing, and healthcare. This powerful integration of real-time data and advanced analytics in healthcare opens new possibilities for predictive care, early diagnosis, and personalized treatment plans.

DTs in healthcare encompass a wide range of applications, including predictive maintenance similar to engineering practices and early diagnosis of diseases [11]. The DT technology has evolved from its initial roots in engineering and aerospace to become a key player in personalized treatment, accurate diagnoses, and the development of simulation models for surgical operations. In this context, DTs serve as virtual representations of patients, offering an innovative approach to health promotion, heightened control over patient health, and improved efficiency in healthcare operations. These DTs can continuously monitor patient health in real—time and adapt their models to predict future health trends, offering doctors and healthcare providers a dynamic and personalized way to treat patients based on both historical and current medical data [12,13]. DTs play a fundamental role in reducing medical errors, thus avoiding misdiagnoses and incorrect treatments, which can have severe consequences [13]. By integrating DT technology with healthcare systems, intelligent and context-aware systems capable of personalized patient care and monitoring can be developed. They can simulate complex medical procedures such as vascular bypass operations in atherosclerosis patients or facilitate the planning of organ transplantations, acting as a bridge between engineering and healthcare practices [13–15]. Furthermore, DT technology can help prevent failures in medical device operation and misinterpretation of medical findings, which are critical to avoiding misdiagnoses and incorrect treatments [13]. Moreover, it has accelerated drug discovery and enabled the creation of biologically detailed digital reconstructions of organs. These "living models" adapt to changes based on data collected from their real-life counterparts, marking a significant advancement from their first use in medical imaging in the mid-1990s.

As DT technology becomes increasingly integrated into healthcare systems, it has sparked discussions about its socio-ethical implications, particularly regarding privacy and individuality in patient care. Questions have arisen regarding whether DTs will worsen or alleviate concerns about privacy and individuality [16]. The healthcare industry's significant investment in DTs reflects its potential impact and importance. Given these concerns, a comprehensive, multi-phase framework is necessary to fully realize DTs' benefits. This framework combines IoT, data analytics, and machine learning (ML) to create accurate virtual replicas of patients, enhancing healthcare professionals' abilities to monitor and improve patient health. It involves various phases such as processing and prediction, monitoring and correction, and comparison, each aimed at improving different aspects of healthcare operations, thanks to various technologies such as sensors, high-speed communication, cloud computing, and AI. The DT is then utilized to predict diagnosis, prognosis, and therapeutic intervention efficacy and optimization [13,17,18]. Fig. 1 illustrates the bidirectional nature of DT systems in healthcare, distinguishing them from conventional one-way predictive models. The process begins with biological data collection from patients across multiple scales (body, organs, tissue, cellular, subcellular levels), which feeds into computational models leveraging AI and machine learning algorithms. These models generate patient-specific digital twins that produce clinical predictions for diagnosis, prognosis, and therapy optimization. The clinical interventions that are informed by these predictions include medication adjustments, treatment modifications, surgical planning, and lifestyle recommendations that modify the patient's biological state. The updated patient data continuously refines the digital twin through real-time monitoring, creating a closed-loop system that enables adaptive, personalized healthcare delivery [13,19].

This study explores the potential applications of DTs in IoT-AI healthcare and their ability to improve patient outcomes and enhance healthcare delivery. The primary research gap and the key contributions of this work are summarized in Table 1. The study highlights areas where DT technology can significantly affect, such as personalized and precision medicine, simulation models for operations and interventions, drug discovery, and the development of biologically detailed brain or heart digital reconstructions. However, the successful implementation of DT technology in healthcare also presents significant challenges, such as privacy concerns, ethical issues, and the need for robust data

Table 1
Statement of significance.

Problem	What is already known	What this paper adds
DT technology faces challenges in healthcare	Existing literature discusses DT applications in healthcare, focusing on real-time monitoring, predictive diagnostics, and simulation models. However, the research lacks a structured review addressing categorization, security concerns, and regulatory issues.	This paper systematically categorizes DTs in IoT-AI healthcare, identifies key components, highlights security and ethical challenges, and suggests future research directions for seamless integration.

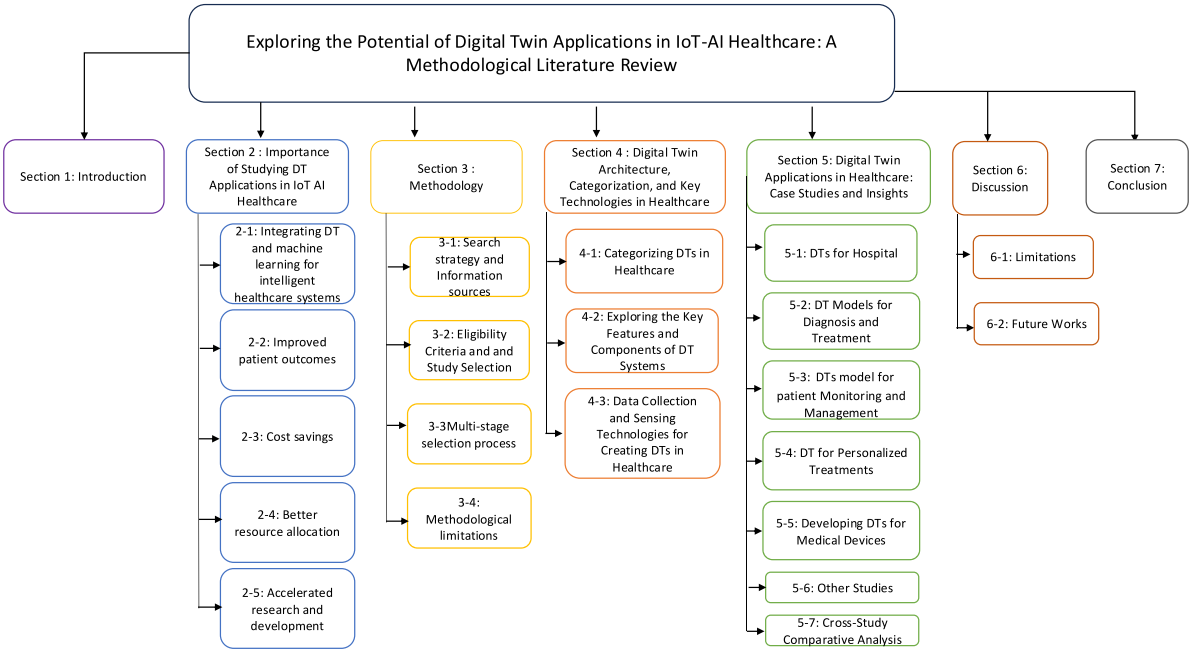


Fig. 2. Paper structure.

security. The findings of this review indicate that utilizing DTs in IoT-AI healthcare has the potential to revolutionize healthcare delivery and enhance patient outcomes. However, successful implementation of this technology is contingent upon addressing the key challenges in the process. Hence, the research question, *"How can the use of DTs in IoT-AI healthcare improve patient outcomes and enhance healthcare delivery, and what key challenges must researchers address for successful implementation?"* remains relevant and pressing for future research. This paper reviews the applications of DTs in IoT-AI healthcare, their benefits in improving patient outcomes, and the key challenges that must be addressed (see Table 1).

The rest of the paper is organized as follows. Section 2 reviews the importance of studying DT technology in healthcare, focusing on its integration with IoT and AI. Section 3 covers the methodology, including search strategy, eligibility criteria, and multi-stage selection process. Section 4 presents DT architecture, categorization, key features, and data collection technologies. Section 5 presents case studies of DT applications across various healthcare domains. Section 6 discusses the foundational challenges, limitations, and future research directions, and Section 7 concludes with a summary of key findings and pathways for clinical translation. Fig. 2 presents a detailed structure of the paper.

2. Importance of studying DT applications in IoT AI healthcare

Many academic and industrial scientists consider using data-driven AI DTs as an alternative to mathematics modeling to improve diagnosis and prognosis and optimize treatments [9]. However, it is essential to note that combining AI and mechanistic approaches is the most effective. Mathematics modeling can provide crucial constraints for training AI algorithms. In this respect, AI can aid Mathematics modeling by assisting in parameter identification, model structure evaluation, simulation acceleration, and continuous model refinement by comparing model predictions with observations. This synergy between AI and traditional mathematical modeling ensures a more robust and comprehensive healthcare solution [20]. Hence, integrating DTs with AI and mechanistic approaches can enhance the accuracy and efficiency of healthcare systems [21]. The study of DT applications within IoT and AI is gaining significant importance in healthcare as these technologies progressively transform how healthcare services are delivered. Hence, exploring the potential of DTs in healthcare is essential to unlocking new possibilities for patient care and operational efficiency. This section explains why studying DT applications in IoT AI healthcare is essential.

2.1. Integrating DT and machine learning for intelligent healthcare systems

Integrating DTs with ML represents a major advancement in healthcare technology. This combination allows for more accurate diagnoses and effective treatments, as seen in developing a heart disease diagnosis system by Elayan et al. [22]. To diagnose heart disease, they developed a context-aware healthcare system utilizing DTs and machine learning, including an ECG heart rhythms classifier model. The ability of these systems to continuously learn and adapt to new data enhances their effectiveness over time, making them invaluable tools in modern healthcare. This integration is particularly important for managing chronic conditions, where continuous monitoring and adaptation of treatment plans can significantly improve patient outcomes. Moreover, by leveraging large datasets, DTs combined with ML can identify subtle patterns in patient data, facilitating early detection of diseases [23].

2.2. Improved patient outcomes

DT technology visually represents a patient's physical self, allowing healthcare providers to create personalized treatment plans and predict

potential health issues to improve patient outcomes. Two key examples of this technology's potential in healthcare include the development of DTs for the human immune system and the use of cardio twins for ischemic heart disease [19].

The development of immune DT technology enables healthcare providers to simulate and analyze various treatment options, leading to highly personalized and effective treatment plans. This technology can also predict and prevent potential health complications, thereby improving patient outcomes. Authors in [24] discussed the development of immune DT technology for biomedical and clinical applications, which presents formidable scientific and technical challenges. They suggested the formation of a consortium for predictive immunology (CPI) that would bring together researchers from various fields to enable the development of immune DTs (IDTs). However, the challenges can be significant and require novel measurement technologies, major scientific discoveries, and technological developments to achieve immune DTs that are robust and accurate enough to approach industrial standards [24].

Similarly, in the case of ischemic heart disease, the continuously updating DT model allows for real-time patient data monitoring, ensuring that treatments remain effective over time by adapting to changes in the patient's condition [25]. Case studies of the cardio twin for ischemic heart disease exemplify how this technology can address complex health issues with precision and efficiency. Authors in [26] introduced Cardio Twin, an architecture based on a DT of the human heart that utilizes a convolutional neural network (CNN) to detect ischemic heart disease (IHD). The developed system can classify non-myocardial and myocardial conditions on edge, making it a valuable tool for IHD diagnosis. The study used the Physikalisch-Technische Bundesanstalt (PTB) diagnostic ECG database from Physio Bank with data samples from 200 people. The implemented model achieved 85.77% accuracy and used 4.8 s for each sample classification, demonstrating that technology can fully support demanding processes such as DTs on the edge [26].

2.3. Cost savings

Cost reduction is an important benefit of DTs in healthcare. DT technology can help healthcare providers identify potential problems before they occur, reducing costly treatments and hospitalizations. DTs can reduce the need for expensive treatments and hospitalizations by predicting health issues before they become severe. In addition to early interventions, DTs streamline operations, offering significant cost-saving opportunities [27]. Some of the benefits of cost reduction in healthcare through DTs are as follows [28]:

- Timely maintenance of equipment and optimized workflow within healthcare institutions can save costs.
- Faster drug development can increase investment opportunities and reduce medicine production costs.
- Shorter treatment periods and error avoidance in diagnosis and treatment can reduce costs.
- Patient typing and self-diagnosis can address less urgent cases outside the physician's office, reducing costs.

DTs can speed up processes requiring more human intervention and raw materials in vivo, reducing costs [29]. The second way to reduce costs is by reducing the time-to-market and cost-to-market values of specific drugs, which can increase investment opportunities and drive the price of medicine production. However, although the potential of the DTs for cost reduction is significant, conflicting views exist on whether it will increase or reduce the overall disease burden. The cost of developing and maintaining models can be high, and it is essential not to overlook the energy consumption associated with computing solutions [30].

2.4. Better resource allocation

The study of DT technology in healthcare is vital for improving resource allocation by accurately predicting where resources are most needed, enabling healthcare providers to optimize personnel, equipment, and facilities [16,31]. By predicting demand, DTs help reduce waste and ensure critical resources are available when needed [13]. For instance, utilizing data across multiple scales, from the subcellular level to the whole body, is essential in constructing DTs for specific health conditions such as viral infections, illustrating the technology's potential to integrate various data types for comprehensive patient care. The current global health crisis has presented numerous challenges for health systems [13], underscoring the need for a viral-infection DT. This DT would be an integration platform for high-dimensional data, helping with evidence-based personalized treatment. Repurposing the immune response portion of the viral DT, which is crucial for diseases such as cancer, where the immune system plays a significant role, could quickly adapt the technology for other applications. Combining DTs with disease-specific twins already in development could lead to more comprehensive models, which in turn could enhance the resilience of the healthcare system and enable a rapid response to emerging viral pathogens. To achieve this, AI and mechanistic modeling play a crucial role in developing and refining these DTs by providing important constraints for AI algorithms assisting in parameter identification, model structure evaluation, and simulation acceleration. As a result, adopting DTs holds significant potential to transform the healthcare industry, particularly in addressing viral infections [32].

2.5. Accelerated research and development

The accelerating role of DT technologies in healthcare research and development is increasingly evident. By enabling in-silico experimentation, virtual clinical trials, and continuous treatment optimization, DT frameworks can substantially reduce both the cost and duration of traditional research cycles. This capability accelerates innovation and supports earlier identification of potential safety or performance issues in therapeutic design and validation [33]. Fig. 3 illustrates the growth trajectory of research publications on DT applications in health and well-being from 2017 through August 2025, based on our systematic literature search primarily using Google Scholar and major academic databases. The trend demonstrates a pronounced acceleration beginning in 2021, coinciding with the broader integration of IoT, AI, and cloud-based health data ecosystems. Publication activity expanded from fewer than ten studies in 2018 to more than 250 works in 2024, with over 260 already published by mid-2025. This represents an exponential increase in scholarly attention to DT applications in healthcare domains such as personalized medicine, patient monitoring, and digital diagnostics.

It is important to note that these values represent a conservative estimate derived from indexed sources; the true number of DT-related studies across preprint servers, institutional repositories, and industry reports is likely higher. The observed acceleration underscores the rapidly expanding interdisciplinary interest and the transformative potential of DT frameworks in advancing medical research and healthcare innovation. The combined progression of DTs with complementary technologies such as IoT, big data analytics, and machine learning, has been a primary catalyst behind this surge [34,35]. Moreover, applications such as organ-on-a-chip and other virtual modeling techniques offer ethical advantages by reducing reliance on animal testing and enabling precision modeling of patient-specific responses [30,36]. This sustained momentum not only accelerates therapeutic discovery but also supports the transition toward predictive, preventive, and personalized healthcare systems. Global spending trends reinforce this trajectory, with IoT in healthcare projected to reach USD 188.2 billion by 2025, an annual growth rate exceeding 21% [37].

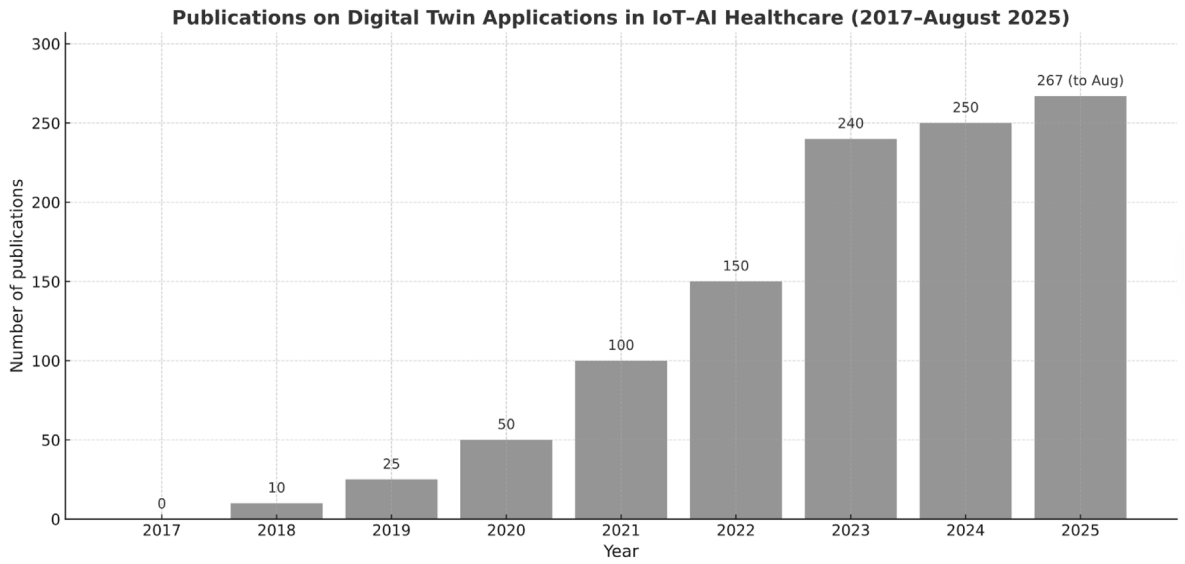


Fig. 3. Growth of research publications on Digital Twin applications in health and well-being (2017–August 2025), based on a systematic literature search. The trend highlights a marked acceleration in research activity beginning in 2021, reflecting the increasing adoption of AI–IoT–driven DT frameworks across healthcare domains.

3. Methodology

This section presents a comprehensive account of the systematic methodological approach employed to identify, select, extract, and synthesize literature on DT applications in IoT-AI-enabled healthcare systems. The review protocol adheres to PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) principles to ensure transparency, reproducibility, and comprehensiveness. The methodology encompasses six primary phases: (i) protocol development and search strategy formulation, (ii) comprehensive database searching and record identification, (iii) multi-stage screening and eligibility assessment, (iv) quality appraisal, (v) systematic data extraction and coding, and (vi) thematic synthesis with quantitative meta-analysis of limitations.

3.1. Search strategy and information sources

A comprehensive literature search was performed across six major academic databases selected for their comprehensive indexing of biomedical, engineering, and computer science literature: PubMed/MEDLINE (biomedical and clinical literature), IEEE Xplore Digital Library (engineering and technology publications), Scopus (multidisciplinary abstract and citation database), Web of Science (cross-disciplinary research), ACM Digital Library (computing and information technology), and ScienceDirect (full-text scientific database). The temporal scope covered publications from 2017 through 2025.

3.1.1. Boolean search string and query construction

The core Boolean search strategy combined three fundamental concept groups to balance sensitivity (capturing all potentially relevant studies) with specificity (minimizing irrelevant retrievals):

```
("Digital Twin*" OR "Digital Twins" OR "Virtual Twin*")
AND
("Healthcare" OR "Health" OR "Medical" OR "Clinical" OR
"Patient*" OR "Hospital*")
AND
("IoT" OR "Internet of Things" OR "AI" OR "Artificial
Intelligence" OR "Machine Learning" OR "Deep Learning")
```

The use of truncation symbols (*) ensured capture of plural forms and variations (e.g., "Digital Twin" and "Digital Twins"). Database-specific search strategies were adapted to accommodate variations in syntax, controlled vocabularies, and indexing practices.

3.1.2. Supplementary targeted searches

To ensure comprehensive coverage of specific application domains and emerging technologies, supplementary targeted searches were conducted using domain-specific terminology. These queries were supplemented with targeted searches on:

- Personalized Medicine: Patient-specific modeling and precision therapy applications
- Predictive Healthcare: Prognostic modeling and early detection systems
- Patient Monitoring: Continuous monitoring and real-time surveillance implementations
- Medical Devices: Implantable devices, prosthetics, and cardiac device integration
- Drug Discovery: Pharmaceutical development and clinical trial optimization
- Electronic Health Records: EHR/EMR integration and health information systems

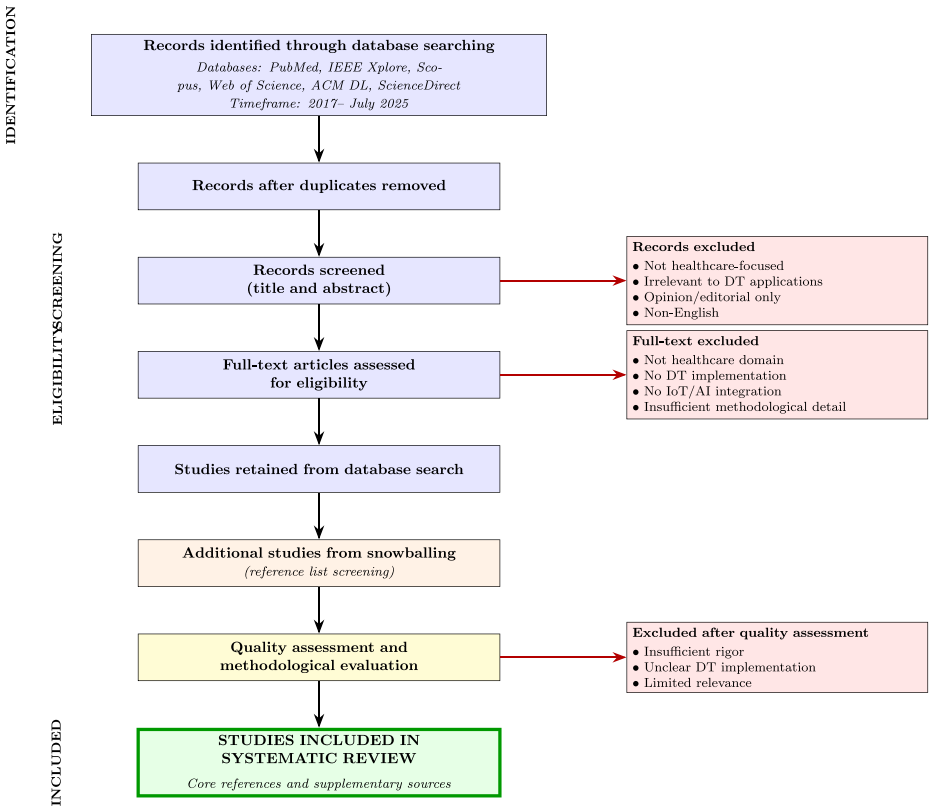
These supplementary searches ensured capture of studies employing domain-specific terminology that might not explicitly use "Digital Twin" nomenclature in titles or abstracts.

3.2. Eligibility criteria and study selection

Clear, predefined inclusion and exclusion criteria were established to ensure systematic and reproducible study selection, applied hierarchically during the multi-stage screening process.

Inclusion criteria Studies were eligible for inclusion if they satisfied the following conditions:

1. Publication Type: Peer-reviewed journal articles, conference proceedings papers, and technical reports
2. Temporal Coverage: Published between 2017 and 2025
3. Language: Full text available in English



PRISMA flowchart for the literature selection process .

Fig. 4. PRISMA flowchart illustrating the literature selection process for Digital Twin applications in IoT–AI healthcare (2017–2025).

4. Healthcare Focus: Studies that (i) addressed healthcare DT applications, (ii) described architectures, frameworks, or implementations in medical settings, (iii) integrated IoT/AI/ML technologies, or (iv) discussed challenges, ethics, or regulatory considerations
5. Technical Content: Provided substantive technical detail regarding DT implementation, validation, or deployment
6. Accessibility: Full text retrievable through institutional access or open-access repositories

Foundational datasets and databases (e.g., MIMIC-III/IV, PhysioNet) were included as established public clinical databases when they represented essential infrastructure for DT development and validation, representing an exception to the peer-review requirement due to their critical role in the field.

Exclusion criteria Studies were excluded if they met any of the following conditions:

1. Non-Healthcare Focus: Studies focused on non-healthcare DT domains (e.g., manufacturing, aerospace, construction) without direct medical relevance
2. Insufficient Technical Depth: Opinion pieces, editorials, or commentaries lacking technical detail
3. Duplicates: Duplicate publications reporting identical findings
4. Methodological Opacity: Studies without clear methodology descriptions
5. Non-Peer-Reviewed: Sources lacking peer review (except established public databases as noted above)
6. Inaccessibility: Full text unavailable despite reasonable retrieval efforts

3.3. Multi-stage selection process

The study selection process followed a systematic, multi-stage approach adapted from PRISMA guidelines, involving progressive refinement through increasingly stringent evaluation. The entire process is illustrated in Fig. 4.

Stage 1: Database search and record identification Database searches were executed between 2015 and 2025, compiling results using reference management software to facilitate deduplication and screening.

Stage 2: Deduplication Duplicate records were identified and removed using a combination of automated algorithms (based on title, author, year, and DOI matching) and manual verification to identify near-duplicates with minor metadata variations.

Stage 3: Title and abstract screening Two reviewers independently screened titles and abstracts of all deduplicated records against the predefined inclusion and exclusion criteria. Studies were conservatively classified to minimize premature exclusion of potentially relevant studies. This screening process resulted in 95 records advancing to full-text assessment.

Stage 4: Full-text eligibility assessment Full-text versions of all potentially eligible studies were retrieved and subjected to detailed eligibility assessment by two independent reviewers. Studies were systematically evaluated against each inclusion and exclusion criterion, with specific reasons documented for excluded studies. Common exclusion reasons included: insufficient technical detail (focus on conceptual discussions without implementation), lack of direct DT implementation, primary focus on non-healthcare applications, and full-text unavailability.

Stage 5: Reference snowballing To identify additional relevant studies not captured by database searches, the reference lists of the 83 selected core articles were manually examined. This snowballing process identified 12 additional articles meeting all inclusion criteria, bringing the total corpus to 95 included studies (83 core references + 12 from snowballing).

Stage 6: Quality assessment All candidate studies underwent quality appraisal considering methodological rigor, clarity of DT implementation, and relevance to the research questions. This quality assessment process resulted in 107 studies being excluded due to insufficient methodological quality, unclear implementation details, or limited relevance, yielding core references meeting all quality criteria.

3.4. Methodological limitations

Despite rigorous protocol adherence, several methodological limitations warrant acknowledgment:

Language and publication bias Restriction to English-language publications may exclude relevant work published in other languages, particularly from non-English-speaking regions with active DT research programs. Additionally, focus on peer-reviewed publications creates potential publication bias toward positive findings.

Database coverage While six major databases were searched comprehensively, some relevant studies indexed exclusively in specialized or regional databases may have been missed. Proprietary industry implementations and unpublished implementations represent a potential gap in coverage.

Terminology heterogeneity Inconsistent terminology poses challenges for comprehensive retrieval. Some relevant studies may use alternative terms (“virtual patient models”, “cyber–physical healthcare systems”, “personalized simulation platforms”) without explicitly using “Digital Twin” terminology, potentially leading to missed studies indexed under alternative terminology.

Gray literature and proprietary systems The review has partial coverage of gray/proprietary implementations, as many commercial DT systems may not be described in peer-reviewed literature, potentially underrepresenting cutting-edge industry developments.

Notwithstanding these limitations, the systematic, transparent, protocolized multi-stage process, including explicit inclusion/exclusion criteria, dual independent screening, standardized data extraction, quality assessment, and innovative quantitative limitations meta-analysis, supports a comprehensive and reproducible overview of DT applications in IoT–AI healthcare.

4. Digital twin architecture, categorization, and key technologies in healthcare

The DT technology is revolutionizing healthcare by creating virtual replicas of physical systems, enabling more accurate and predictive healthcare delivery. Through the integration of IoT and AI, DTs offer real-time data processing, simulation models, and predictive analysis, which can significantly enhance patient care, streamline healthcare operations, and improve clinical decision-making [38,39]. In this section, we explore the categorization of DTs in healthcare, their key features and components, and the technologies involved in data collection and sensing. These insights provide a foundation for understanding how DTs can transform healthcare delivery and the challenges that must be addressed to ensure successful implementation.

4.1. Categorizing DTs in healthcare

DTs in healthcare have the potential to revolutionize patient care by providing a virtual replica of a patient or a healthcare system. However, with various applications and underlying technologies used to build and operate them, DTs can be challenging to understand and implement in healthcare systems [27]. To ensure the effective and efficient use of DTs and to weigh the advantages and disadvantages of proposed models, it is essential to consider different approaches for categorizing them. In this context, *the categorization based on the application* can help healthcare providers find which DT category is most appropriate for their needs [40]. In contrast, *categorization based on technology* can supply insights into the tools and technologies used to build and operate DTs. DTs can utilize IoT and ML-based strategies by collecting real-time data from physical systems and using ML algorithms to analyze and interpret the data. This approach is beneficial for real-time monitoring and control applications, such as manufacturing or energy sectors [41].

On the other hand, *categorization based on real-time functions* can help identify the level of interaction between the DT and its physical counterpart, which is decisive for developing accurate and reliable DTs [42]. By understanding and utilizing these categorization approaches, healthcare providers can better leverage DTs to improve patient outcomes and enhance the healthcare system [41]. Categorizing DTs based on real-time functions uses real-time functions and categorizes DTs based on the degree of real-time interaction between the DT and its physical counterpart. DTs can be active, semi-active, or passive; continuous real-time data collected from the physical twin updates active DTs, while semi-active DTs collect time-varying data that analysts examine after data collection. In contrast to active DTs that constantly update data from their real-life counterparts, passive DTs rely on measurements taken from a physical twin that may include some modeling assumptions [38]. This approach is vital for developing correct and reliable DTs [41]. Authors in [43] developed a semi-active DT model to detect carotid artery stenosis noninvasively. They used a time-dependent face video of a subject to calculate the head vibration before comparing the in vivo value to computationally generated data. This approach allowed them to determine the severity of carotid artery stenosis. While this is not an active DT model, it is a step toward creating a more comprehensive DT model. In this context, the advancement of ML algorithms and computational methods can lead to the development of more sophisticated models, enabling more accurate and real-time monitoring of physical systems. Table 2 reveals three complementary categorization frameworks, each serving distinct roles in healthcare DT implementation. **Application-based categorization** operates at the strategic level, enabling clinical leadership to align DT capabilities with care delivery goals without requiring deep technical expertise. **Technology-based categorization** addresses infrastructure selection, directly impacting performance, scalability, and costs—critical for technical teams designing DT systems. **Real-time function-based categorization** defines interaction dynamics: active DTs enable time-critical interventions (e.g., arrhythmia detection), semi-active DTs balance efficiency with clinical utility (e.g., periodic assessments [43]), and passive DTs excel in planning applications. Effective implementation requires hierarchical integration: (1) identify objectives using application-based categorization; (2) select infrastructure based on technical requirements; (3) determine interaction levels based on clinical urgency. Critical research gaps include standardized evaluation metrics, hybrid categorization schemes, and cost-benefit analyses to inform evidence-based adoption decisions.

4.2. Exploring the key features and components of DT systems

Digital Twins (DTs) are virtual counterparts of physical objects, systems, or processes that enable real-time monitoring, control, and optimization of their physical referents. In healthcare, DTs can enhance patient monitoring, diagnosis and treatment, disease management, and

Table 2
Comparative analysis of digital twin categorization approaches in healthcare.

Approach	Purpose	Abstraction level	Examples	Representative clinical use cases
Application-Based	Select DT category based on clinical requirements	High	Patient-specific organ DTs; hospital operational DTs; drug discovery models	Personalized treatment planning; resource allocation; drug efficacy prediction [40]
Technology-Based	Classify by computational infrastructure and data methods	Medium	IoT sensor networks; ML/AI diagnostic systems; cloud/edge platforms	ICU monitoring; automated diagnostics; multi-site care coordination [41]
Real-Time Function	Categorize by data exchange cadence and bidirectional coupling	Low	*Active:* continuous cardiac monitoring; *Semi-active:* carotid stenosis detection; *Passive:* surgical planning	*Active:* ICU monitoring, insulin delivery; *Semi-active:* periodic assessments; *Passive:* pre-op simulation [38]

prevention, while optimizing clinical operations under strict privacy and safety constraints. To deliver these capabilities at scale, contemporary DTs adopt a *system-oriented*, layered architecture comprising: (i) a *Physical/IoT Sensing* layer (wearables, clinical devices, WBANs); (ii) a *Communication* layer (edge/cloud transport, lightweight publish–subscribe); (iii) a *Cyber/Digital* layer (twin graphs, time-series stores, analytics engines); and (iv) a *Cognitive* layer (AI/ML/decision modules). Cloud–edge implementations with explicit optimization components have demonstrated concrete operational gains in clinical telemetry—e.g., a Pyomo-based DT framework reduced telemetry latency by **32%** and improved response time by **52%** for multimodal vital monitoring (SpO_2, HR, BT), while maintaining scalable sensor integration via a twin-graph platform [44]. Hybrid DTs that fuse real-time sensing with historical EHR (e.g., MIMIC-III) report strong predictive performance (overall accuracy ~98.9%, real-time ~95.4%) when coupled with end-to-end encryption and regulatory alignment (HIPAA/GDPR), underscoring the role of *secure-by-design* architecture in clinical adoption [45].

This layered architecture extends naturally to more complex *five-dimensional* frameworks that explicitly model: (1) the physical entity (patients, clinical devices), (2) virtual models (computational replicas), (3) bidirectional connection mechanisms (data flow, control feedback), (4) digital twin data repositories (time-series, knowledge graphs), and (5) service layers (clinical decision support, workflow integration) [46]. The explicit separation of data governance (dimension 4) from service delivery (dimension 5) proves particularly valuable in healthcare, where regulatory requirements (HIPAA, GDPR) demand clear data lineage and access control distinct from clinical application logic.

Beyond centralized pipelines, *federated digital twins* (FDTs) use federated/split learning to collaboratively train across departments without raw-data pooling, while optimizing *timeliness* via the Age of Information (AoI). Specifically, organ-specific DTs (e.g., cardiology, neurology, radiology) function as independent training nodes where feature extraction occurs locally on departmental data, with only learned representations transmitted to a central aggregation server [47]. This split-learning architecture ensures that "raw medical data never leaves its origin facility" while enabling collaborative model improvement across institutions, which is a critical requirement for multi-site clinical research and rare disease modeling where no single institution holds sufficient cases [48]. Deep reinforcement learning (DRL)–based scheduling of data/model refresh has been shown to keep diagnostic models up to date under emergency constraints, achieving 50% AoI reduction (0.6 s vs. 1.0 s) compared to baseline methods—a meaningful improvement when "diagnostic time must be minimized to seconds rather than minutes" in acute scenarios like myocardial infarction within the critical 60-minute therapeutic window [47]. These architectural patterns, including edge offloading, federated learning, and explicit timeliness optimization, jointly address the core healthcare requirements of latency, privacy, scalability, and resilience [44,47].

However, current FDT implementations face notable challenges in "managing diverse data distributions, adapting to ever-changing medical environments, and meeting the demands of real-time decision-making", particularly when balancing model performance with system efficiency across heterogeneous hospital IT infrastructures [48]. The integration of multimodal clinical data like structured EHR records, unstructured clinical notes, medical imaging, laboratory results, and continuous wearable sensor streams remains technically complex, with "inconsistencies in the data, such as missing values, conflicting records, or outdated information" undermining predictive accuracy [46]. Robust preprocessing pipelines and domain-adapted fine-tuning are necessary but not yet standardized across healthcare DT deployments.

Finally, adoption hinges on *human factors*: DT front ends must provide clear visualizations and responsive interactions to fit clinical workflows; interpretable, auditable decision traces increase clinician trust; and robust privacy (E2EE, RBAC) improves patient acceptance by aligning security posture with perceived benefit [44,45]. Natural language interfaces represent a promising approach to reduce workflow friction, allowing clinicians to "design experiments directly using language, significantly lowering the barrier" without requiring programming expertise or complex query syntax [46]. Yet interactivity challenges persist. Current systems often lack the conversational depth to handle clinician follow-up questions, and LLM verbosity — where overly detailed responses undermine clarity and overwhelm end users — remains a recognized usability barrier that warrants targeted interface design attention [46].

4.2.1. AI-enhanced cognitive intelligence in DTs

The cognitive layer determines how a DT learns from data, reasons under uncertainty, and closes the loop with safe, timely actions. Several emergent AI paradigms strengthen this layer for healthcare:

LLMs and agentic AI. Large Language Models (e.g., GPT-4–class) can operate as *cognitive agents* inside DTs to fuse multimodal signals and medical knowledge, generate natural-language rationales, and coordinate decisions. Agentic patterns (*memory–reflection–planning*) provide traceable, intervention-ready reasoning chains that support clinician oversight and safety processes [46,49]. This improves semantic interoperability across DT components and makes complex model behavior auditable in practice [49].

Functionally, LLMs serve four distinct roles within healthcare DTs: (1) *data collectors* that retrieve relevant information from heterogeneous clinical sources (literature, guidelines, EHRs) or generate synthetic training data to address rare-condition data scarcity; (2) *information processors* that extract task-specific representations from noisy multimodal inputs and distill actionable insights; (3) *decision-makers* that simulate patient trajectories, design computational experiments (e.g., treatment protocol variations), and analyze results to derive optimal strategies; and (4) *controllers* that execute recommendations via API integration or present natural-language explanations to clinicians for human-in-the-loop validation [46]. For example, LLMs can automatically generate and execute experimental scenarios in a DT environment by converting natural language instructions into executable code, analyze the resulting data, and recommend personalized treatment plans—demonstrating "the potential to become valuable tools for designing and generating virtual scenarios" for treatment optimization [46].

The explainability advantages are particularly salient for clinical trust. By "emulating human-like thought processes, LLMs can generate intermediate reasoning steps that connect input data to final predictions", enabling chain-of-thought explanations like: *"Based on patient's elevated cardiac enzymes (troponin=0.8 ng/mL) and ST-segment elevation in leads II, III, aVF, the DT predicts 85% probability of inferior wall MI. Recommended immediate catheterization based on ACC/AHA guidelines and similar cases in training data"* [46]. Integration with domain-specific knowledge bases further strengthens credibility by grounding recommendations in established clinical guidelines rather than opaque learned patterns alone.

Critical limitations must be acknowledged. LLM "hallucinations, insufficient reasoning abilities, and knowledge obsolescence persist. These issues can lead to incorrect predictions and, in some cases, severe system failures" [46]. For safety-critical medical decisions, rigorous validation protocols that include systematic testing against clinical ground truth, expert review of generated recommendations, and fail-safe mechanisms for low-confidence predictions are essential prior to deployment. The "unpredictability of LLM behavior and the potential for adversarial attacks pose significant safety risks, particularly in high-stakes domains such as healthcare" [46], demanding defense-in-depth strategies including input sanitization, output verification, and mandatory clinician oversight for high-risk interventions.

Multi-agent systems (MAS). Healthcare DT ecosystems span devices, wards, and facilities. *MAS* coordinate distributed twin agents to share state, negotiate priorities (e.g., triage, imaging queues), and execute collaborative policies. LLM-driven MAS add interpretable, language-level coordination and richer global situational awareness across agents [46,49]. Concretely, natural language communication between agents "reduces the need for rigid communication protocols and enhances explainability" [48]. For instance, enabling seamless handoffs between cardiology, neurology, and radiology DTs during multi-organ patient monitoring, or coordinating diagnostic and treatment-planning agents with transparent, auditable decision rationales. This architecture proves particularly valuable for complex cases requiring multidisciplinary input, where rigid API-based coordination would impose excessive integration overhead and obscure clinical reasoning chains.

Reinforcement learning (RL). Reinforcement learning provides the decision-making backbone of intelligent DT systems by enabling adaptive optimization under uncertainty. Recent healthcare applications formulate treatment planning, scheduling, and patient-state prediction as Markov Decision Processes (MDPs), where the state encodes real-time DT representations, available resources, and clinical context, while actions correspond to diagnostic, therapeutic, or scheduling interventions. The reward function combines clinical benefit, operational efficiency, and fairness objectives to ensure ethical optimization [50].

In surgical scheduling, RL integrated with patient-specific DTs achieved a 55% reduction in average waiting time, 42% decrease in clinical risk, and 16% improvement in operating room utilization, without increasing capacity, demonstrating the feasibility of learning-based prioritization in constrained healthcare environments [50]. The agent continuously refines its policy through simulation feedback, balancing fairness penalties with efficiency rewards to avoid bias and resource underuse. Complementary studies employing deep reinforcement learning (DRL) for patient-twin modeling have validated its diagnostic power. A deep Q-learning architecture trained on multimodal health data (EHRs, sensors, imaging) accurately classified patient health status and forecasted deterioration risks, achieving high accuracy, completeness, and F-measure in simulation trials [51]. The digital twin updates its predictions iteratively as new patient data arrive, demonstrating dynamic adaptability and improved diagnostic reliability. Together, these findings show that RL-enabled DTs function as adaptive and ethically aligned frameworks that personalize decisions, reduce latency, and balance fairness, safety, and efficiency, advancing intelligent clinical decision support.

Federated/split learning (FL/SL). To preserve privacy while exploiting multi-institutional diversity, FL/SL enable collaborative improvement of DT models without centralizing PHI; combined with AoI-aware refresh, FDTs support rapid adaptation in emergencies [47]. The split-learning paradigm partitions neural networks such that "feature extraction occurs at worker nodes while classification is performed at the server, enabling privacy-preserving collaborative learning" [48]. Each hospital's DT processes local patient data through its feature-extraction layers, transmits only the resulting embeddings (not raw data) to a federated server, receives classification results and gradient updates, and updates its local model—maintaining HIPAA compliance as "healthcare facilities retain complete ownership and control of their patient data throughout the entire process" [48]. his architecture enables learning from rare diseases and diverse patient populations without violating institutional data-sharing agreements, though challenges remain in handling "heterogeneous hardware capabilities across edge devices" and "dynamically adjusting batch sizes and computational loads according to each device's capacity" [47].

Embodied AI (EA). DTs provide high-fidelity, data-rich virtual environments where *embodied* agents (e.g., rehab robots, surgical assistants) learn perception–action policies safely before deployment, helping bridge the sim-to-real gap in human–robot interaction and clinical robotics [48]. By simulating "human responses and conditions within a virtual space, HDTs [Human Digital Twins] allow robots to learn without posing a direct threat to human safety" during training, particularly critical for emergency intervention scenarios where real-world mistakes could cause patient harm [48]. Applications span surgical assistance (learning haptic feedback and visual-motor coordination in high-fidelity DT surgical theaters) and patient monitoring (embodied agents adapting to fatigue/stress indicators through DT-simulated patient states). However, current embodied AI systems remain limited to small indoor environments because of the high time and cost of scaling to larger, more complex settings, creating a barrier to hospital-wide deployment that demands advances in simulation scalability and transfer learning [48].

Vision-and-language navigation capabilities, while demonstrated primarily in manufacturing contexts, offer translatable potential for hospital logistics. Natural language–instructed autonomous guided vehicles (AGVs) that navigate via LLM-generated Python code and pathfinder algorithms could support medication delivery, equipment transport, and supply chain automation, with the same interface allowing nurses and clinicians to request items "without learning specific system commands" [52]. In addition to 3D point-cloud reconstruction, semantic annotation, and LLM command comprehension, the technical architecture can be applied directly to hospital corridors and clinical spaces, although regulatory compliance remains a challenge for autonomous systems near patients.

Security, explainability, and UX as acceptance levers. Across these paradigms, clinical acceptance is driven by (i) strong, transparent security (E2EE, RBAC, compliance), (ii) interpretable/traceable recommendations
(LLM/agentic reasoning), and (iii) measurable UX gains (reduced latency, clear visual narratives, lower cognitive load) [44,45,49].

From a security perspective, technical privacy preservation, while necessary, proves insufficient alone; patients require clear communication about what data is collected, how it is used, and who can access it. The combination of federated architecture (localized data), end-to-end encryption (protected transmission), and role-based access control (granular permissions) addresses institutional and regulatory requirements, yet patient trust depends equally on transparent consent processes and demonstrated organizational commitment to data stewardship [46,48].

Explainability mechanisms directly counter the "black box" perception that undermines clinician trust in AI-driven recommendations.

LLM-based DTs that "generate natural language explanations of system decision-making by leveraging domain-specific knowledge bases" [46] transform opaque predictions into clinically interpretable rationales. Chain-of-thought reasoning traces enable clinicians to audit the logic connecting patient data to recommendations, identify potential errors (e.g., missing contraindications), and understand model uncertainty and build the confidence necessary for high-stakes medical decisions. However, "challenges persist in managing complex decision models and improving the interactivity of these explanations", particularly when clinicians pose follow-up questions or request alternative scenarios [46].

UX considerations determine whether DTs integrate smoothly into clinical workflows or create additional friction. Natural language interfaces "significantly lower the barrier for applying predictive digital twin systems" by eliminating programming requirements [46], yet training demands persist, clinicians must still learn when to trust versus override DT recommendations, how to interpret confidence intervals, and what actions to take when system behavior appears anomalous. Visual feedback mechanisms (e.g., real-time patient state evolution, treatment progress trajectories) and responsive interfaces reduce cognitive load, though optimal design patterns remain an active research area. The demonstrated latency improvements (32% reduction in telemetry delay, 52% faster response times) [44] represent measurable UX benefits, but these simulation-validated metrics require real-world clinical validation to confirm perceived benefit and workflow integration.

4.2.2. Edge-cloud orchestration and computational architecture for healthcare DTs

The computational demands of real-time healthcare DTs necessitate hybrid edge-cloud architectures that strategically partition workloads between resource-constrained bedside devices and scalable cloud infrastructure. This architectural decision fundamentally shapes system latency, energy consumption, data privacy, and clinical utility [44,45,47].

- **Edge-Dominant Architectures** prioritize ultra-low latency for time-critical interventions, achieving sub-200ms response times through local processing of real-time signal preprocessing, lightweight anomaly detection, and immediate alert generation [44,53]. However, edge devices face constraints in storage capacity (<32 GB), computational power, and battery life (24–72 h) [54].
- **Cloud-Dominant Architectures** leverage centralized computing for complex tasks including deep learning model training, multi-scale mechanistic simulations (hours per analysis), and population-level analytics [55–58].
- **Hybrid Edge-Cloud Architectures** dynamically distribute workloads based on urgency and complexity. Jameil and Al-Raweshidy [44] demonstrated 32% latency reduction and 52% response improvement through Pyomo-based optimization across edge (vital sign acquisition), fog (medium-complexity analytics), and cloud (centralized training) tiers using twin-graph synchronization.

Latency analysis and performance tradeoffs Latency in healthcare DT systems arises from multi-stage sensing, communication, computation, and visualization processes [45,59]. Cloud-centric frameworks incur substantially higher end-to-end latency than edge-based architectures due to variable network transmission and remote processing delays [47,59]. Clinical implications vary significantly: emergency cardiac care requires sub-second responsiveness within the 60-minute therapeutic window [47,53], while treatment planning applications tolerate hour-scale latencies for increased model fidelity [56,57]. Optimal deployment paradigms must align latency requirements with clinical objectives.

Federated DTs and privacy-preserving edge intelligence Federated learning (FL) enables collaborative DT intelligence without centralized data pooling, addressing HIPAA/GDPR constraints [47,48,60]. Split learning partitions neural networks such that feature extraction occurs locally while classification executes centrally, ensuring raw medical data never leaves origin facilities [48]. Age-of-Information (AoI)-aware scheduling achieves 50% AoI reduction (0.6 s vs 1.0 s) for time-critical applications [47]. However, non-IID data across institutions can degrade performance by 20%+ compared to centralized training [45,47].

Energy efficiency and computational constraints Continuous IoT-based DT operation demands substantial energy for wearables and cloud infrastructure [30,54]. Energy-efficient strategies include model compression (60%–80% reduction with <5% accuracy loss), adaptive sampling (3–5× battery extension), and event-driven processing (40%–70% power savings) [47,54]. Computational bottlenecks include real-time multi-modal fusion ($O(n^2)$ complexity), high-dimensional state spaces (10^4–10^6 variables), and concurrent user scalability (degradation beyond ~50 users) [44,45,55,63–65].

Implementation recommendations Table 3 synthesizes architectural recommendations aligning clinical priorities with technical capabilities. Edge-dominant architectures suit ICU/emergency monitoring requiring sub-second latency [59]; cloud-dominant architectures enable treatment planning tolerating hour-scale computation [55,56]; hybrid edge-cloud balances real-time alerts with longitudinal analytics [44]; federated architectures preserve privacy for multi-site trials despite 20%+ performance degradation under non-IID data [47]. Future directions include adaptive workload migration, neuromorphic edge inference, quantization-aware training, and standardized benchmarking protocols.

4.3. Data collection and sensing technologies for creating DTs in healthcare

Effective DT implementation in healthcare depends critically on multimodal data integration from sensors and clinical systems. This subsection analyzes data acquisition architectures, sensor technologies, clinical research datasets, and persistent integration challenges that determine DT fidelity and clinical utility.

4.3.1. Multimodal data sources and IoT infrastructure

Healthcare DTs integrate three foundational data streams: (i) *structured clinical data* from EHR systems capturing demographics, medical history, diagnoses, treatments, and outcomes [27]; (ii) *medical imaging* (CT, MRI, X-ray) enabling 3D anatomical modeling for diagnosis and surgical planning; and (iii) *continuous physiological monitoring* via wearable devices providing real-time heart rate, blood pressure, oxygen saturation, and glucose levels [66,67].

The Internet of Medical Things (IoMT) extends IoT connectivity through implantable sensors, body-attached wearables, and environment-embedded monitoring systems [54]. Real-time data integration enables proactive intervention through early detection of physiological deterioration, particularly beneficial for chronic disease management where continuous monitoring reduces hospital visits [39]. The global HDT market growth projection from $29.51 billion (2022) to $530 billion (2032) reflects increasing clinical adoption [39].

4.3.2. Sensor technologies and data transmission

Contemporary DT architectures leverage MQTT (Message Queuing Telemetry Transport) protocols for data collection, storing sensor readings in time-series databases enabling real-time monitoring and historical trend analysis [68]. Wearable sensor systems simultaneously capture body temperature (BT), ECG, heart rate (HR), and estimate blood pressure (BP) via Pulse Arrival Time (PAT) from ECG and photoplethysmography (PPG) signals [54].

Low-power wireless transmission via Bluetooth Low Energy (BLE) modules incorporates encryption at both sensor and gateway levels

Table 3
Architectural decision framework for healthcare digital twins.

Clinical application	Recommended architecture	Rationale	Representative studies
ICU/Emergency monitoring	Edge-dominant (with cloud backup)	Sub-second latency required; local processing ensures continuity during network outages	[53,59]
Chronic disease management	Hybrid edge-cloud	Balance between real-time alerts (edge) and long-term trend analysis (cloud)	[44,45,61]
Treatment planning	Cloud-dominant	Complex simulations tolerate hour-scale latency; centralized resources enable sophisticated modeling	[55–57]
Multi-site clinical trials	Federated edge-cloud	Privacy preservation mandatory; distributed learning enables collaboration	[47,48]
Hospital operations	Cloud-based with fog aggregation	Facility-wide optimization benefits from centralized coordination; fog layer reduces WAN traffic	[62,63]

to maintain privacy and security. Narrowband IoT (NBIoT) dominates cellular IoT healthcare applications due to power efficiency suitable for extended wearable device operation [54]. These transmission architectures balance real-time responsiveness with battery constraints and privacy requirements.

4.3.3. Clinical research datasets

Clinical research datasets provide the empirical foundation for DT algorithm development by supplying large-scale, de-identified patient records that enable training and validation without privacy violations or prospective trial requirements. These datasets exhibit three critical characteristics determining DT utility: (i) temporal resolution spanning hours to years for longitudinal modeling, (ii) modality diversity integrating structured EHR data with unstructured clinical notes and physiological waveforms, and (iii) phenotypic heterogeneity capturing population variability essential for generalization. This subsection examines how MIMIC-III/IV architectures and specialized physiological databases instantiate these characteristics while constraining DT performance through inherent biases, limited sample sizes, and single-center data collection.

MIMIC-III architecture and DT applications. The MIMIC-III database provides comprehensive ICU patient data including vital signs, medications, laboratory measurements, provider notes, procedure codes, diagnostic codes, imaging reports, and survival outcomes [69]. Fig. 5 illustrates the multi-stage pipeline: raw clinical data from five ICU departments (MICU, SICU, CCU, CSRU, NICU) undergoes de-identification, date-shifting, and format conversion to produce research-ready datasets.

MIMIC-III's architecture enables three critical DT modeling paradigms. First, structured vital signs and waveform data support real-time cardiac monitoring DTs where continuous ECG, heart rate, and blood pressure streams feed edge-based classification models achieving 85.77% accuracy [26]. Second, longitudinal laboratory measurements enable patient-specific metabolic panel DTs using stacked LSTM architectures to forecast treatment response dynamics [61]. Third, comprehensive demographic and outcome data facilitate population-level hospital operational DTs for resource allocation optimization [62]. The combination of structured EHR data with unstructured clinical notes provides the multimodal training foundation for hybrid cloud-IoT DT frameworks achieving 95%–99% predictive accuracy by fusing MLP and XGBoost models across historical records and real-time sensor streams [45].

MIMIC-IV enhanced granularity. MIMIC-IV extends the foundational architecture with enhanced temporal and spatial resolution: high-resolution ECG data (10-second, 12-lead), emergency department vital signs, DICOM chest X-rays with annotations, comprehensive medication logs, and clinical notes [70] (Fig. 6). This enhanced granularity enables three advanced DT capabilities. First, arrhythmia detection DTs using CNN architectures on edge devices achieve 4.8-second inference times [26]. Second, spatiotemporal cancer treatment response DTs combine longitudinal MRI with mechanistic physiological models to forecast neoadjuvant chemotherapy outcomes in triple-negative breast cancer [55]. Third, federated DT architectures train departmental models (cardiology, neurology, radiology) locally on MIMIC-IV subsets and aggregate learned representations centrally without exposing raw patient data, achieving 50% Age-of-Information reduction for time-critical diagnostics [47].

The explicit data stratification (diagnostic ECG/X-ray, operational ED triage/demographics, treatment medications/procedures, outcome discharge summaries) directly maps to the layered DT architecture (Section 4.2), where physical sensing, communication, cyber/digital, and cognitive AI layers process heterogeneous streams into synchronized patient-specific twins.

Specialized physiological datasets. The MIT-BIH Arrhythmia Database provides 48 half-hour two-channel ambulatory ECG recordings with beat-by-beat cardiologist annotations for arrhythmias including PVCs, atrial fibrillation, and ventricular tachycardia [71,72], serving as benchmark data for arrhythmia detection algorithm development. Gene expression profiling datasets from breast cancer biopsies (e.g., docetaxel resistance studies [73]) enable DT-based virtual tumor replicas for treatment simulation before clinical administration [74], demonstrating domain-specific dataset requirements for personalized therapy optimization.

4.3.4. Data quality, privacy, and integration challenges

Four fundamental barriers constrain effective dataset utilization in healthcare DT systems:

- Data Quality and Heterogeneity: HDT development requires high-fidelity modeling from potentially scarce, biased, and noisy data [39]. Human physiological heterogeneity demands diverse multimodal integration — physiological measurements, environmental factors, behavioral patterns — substantially exceeding conventional mechanical system DT requirements. This heterogeneity creates systematic data quality challenges including missing values, temporal misalignment, and inconsistent sampling rates across modalities.
- Privacy and Security: Individual-level HDT data sensitivity necessitates robust security schemes to prevent ethical and regulatory violations [39]. IoT device proliferation generates overwhelming structured and unstructured data volumes, creating management challenges under resource and time constraints. Without proper auditing and organization, healthcare data becomes redundant, incomplete, or obsolete, undermining DT accuracy and clinical utility.
- Standardization Gaps: Specialized communication standards (ETSI SmartBAN, IEEE802.16.5) for healthcare IoT remain under adopted in device design [54]. Healthcare-specific DT standards must address safety, security, biological diversity, and ethical issues beyond generic IoT frameworks. The absence of standardized protocols for large-scale biomedical dataset collection, storage, and processing across diverse computing platforms hinders deployment scalability [54].

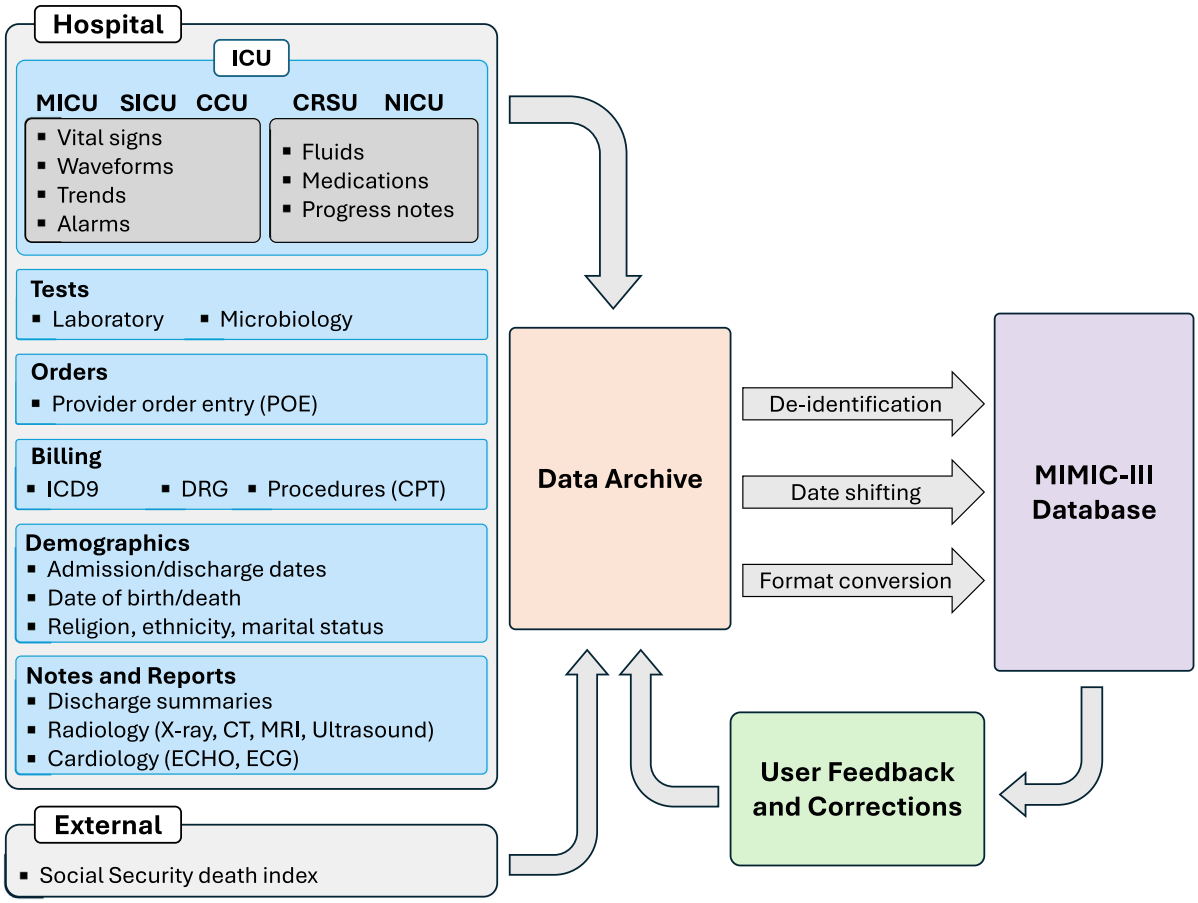


Fig. 5. MIMIC-III database collection process [69].

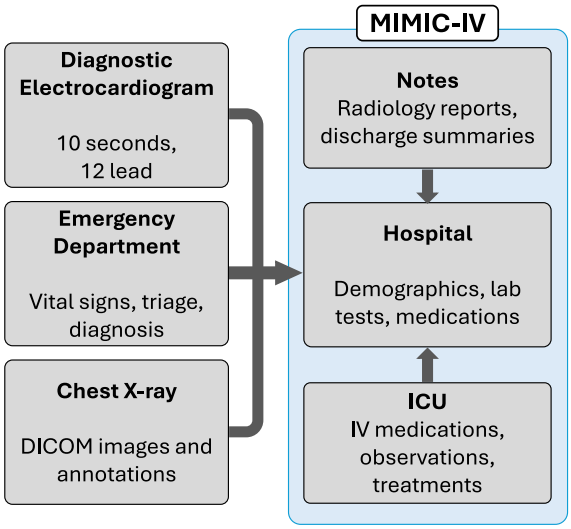


Fig. 6. MIMIC-IV database collection process [70].

- Infrastructure and Computational Demands: Multimodal deep learning models, advanced biosensors, and quantum computing represent future enhancement pathways for real-time monitoring and predictive analytics [54]. However, current infrastructure limitations constrain simultaneous processing of high-frequency physiological signals, medical imaging, and longitudinal EHR data, particularly in resource-constrained clinical environments.

Table 4 synthesizes twelve heterogeneous data modalities for comprehensive healthcare DTs, systematically comparing sensor types, captured information, sampling frequencies, DT applications, and privacy/integration considerations across clinical research datasets, continuous monitoring systems, and specialized physiological measurements [39,54,68–70].

5. Digital twin applications in healthcare: Case studies and insights

DTs in healthcare are cyber–physical surrogates continuously synchronized with their physical counterparts through multimodal data streams such as electronic health records, imaging, and wearable sensors, supported by computational models including mechanistic and machine learning approaches. This section presents representative case studies across four application areas: hospital operations, diagnosis and treatment, patient monitoring and management, and medical devices, emphasizing system architecture, data pipelines, and feedback mechanisms.

5.1. DTs for hospital

Authors in [62] developed a smart hospital system known as "Hospital Intelligent Twins" to enable all-scenario intelligent healthcare. They used Building Information Modeling (BIM) technology to implement this system to support creating and managing digital hospital assets. The system comprised three main components: the hospital smart brain, clinical department DTs, and intelligent wards. The system integrated five intelligent application modules to manage the hospital's overall situation, medical operations, security, firefighting, and

Table 4
Multimodal data sources and sensing technologies for healthcare digital twins.

Data modality	Source/Sensor type	Captured information	Sampling frequency	DT application	Privacy/Integration considerations
Structured Clinical Data	Electronic Health Records (EHR)	Demographics, medical history, diagnoses, treatments, outcomes	Event-triggered	Longitudinal patient modeling, treatment outcome prediction	HIPAA/GDPR compliance, access control, data de-identification
Medical Imaging	CT, MRI, X-ray, Ultrasound, PET	3D anatomical structures, tissue characteristics, pathological features	On-demand	Organ modeling, surgical planning, disease progression tracking	DICOM standards, large file storage, computational cost
Cardiac Monitoring	ECG sensors (wearable, clinical)	Heart rate, rhythm, arrhythmia events, PAT-based BP	Continuous or interval-based	Real-time arrhythmia detection, cardiovascular DT	BLE encryption, battery life, motion artifacts
Continuous Glucose Monitoring	Wearable CGM	Blood glucose levels, trends, variability	1–5 min intervals	Diabetes management DT, insulin optimization	Calibration requirements, sensor drift, replacement cycles
Activity and Fitness Tracking	Smartwatches, fitness bands	Steps, calories, sleep patterns, activity intensity	Continuous to interval	Rehabilitation monitoring, health assessment	User compliance, device placement variability
Blood Pressure Monitoring	Smart cuffs, wearable BP monitors	Systolic/diastolic pressure, pulse pressure	Interval or on-demand	Hypertension management, cardiovascular risk assessment	Calibration, cuff placement errors
Pulse Oximetry	Wearable SpO_2 sensors	Blood oxygen saturation, pulse rate	Continuous	Respiratory monitoring, patient tracking	Perfusion index, skin tone calibration
Body Temperature	Wearable thermometers, smart patches	Core/skin temperature, fever detection	Continuous	Infection monitoring, post-surgical recovery	Sensor placement, ambient temperature compensation
Behavioral Data	Mobile apps, social media	Mood, stress, activity patterns	Event-triggered	Patient engagement, mental health monitoring	User consent, data validity, privacy concerns
Medication Adherence	Smart pillboxes, connected inhalers	Dosage timing, missed doses, usage patterns	Event-triggered	Treatment adherence monitoring, therapy optimization	False positives, autonomy concerns
Genomic Data	Sequencing platforms	Genetic variants, expression profiles	One-time or periodic	Precision medicine DT, drug response prediction	Secure storage, ethical consent, computational burden
Public Research Datasets	MIMIC-III/IV, MIT-BIH, PTB-XL	De-identified ICU records, ECG annotations	Retrospective	DT model training, algorithm validation	Dataset bias, temporal shift, generalization limitations

Note: BLE = Bluetooth Low Energy; DT = Digital Twin; GDPR = General Data Protection Regulation; HIPAA = Health Insurance Portability and Accountability Act.

energy consumption conditions. The clinical department DTs allow for the visualization and simulation of hospital operations, such as analyzing MRI and CT usage, allocating resources, and monitoring patient conditions. As reported, the intelligent ward can promote intelligent interaction among devices in three scenarios: ward, corridor, and nurse station. Moreover, the system provides personalized healthcare by implementing personalized health recommendations, persistent monitoring, and electronic education for personalized care. In another study [75], the authors proposed improving hospital facilities' design and construction to withstand better extreme weather events such as hurricanes, floods, and wildfires. The developed paradigm can help ensure that hospitals can continue functioning during and after such disasters, allowing them to provide critical patient care. However, while the study does not directly address patient diagnosis and treatment, it is still an important consideration, as the ability of hospitals to function during disasters can significantly impact public health. Fig. 7 demonstrates the main steps to create a hospital's DT: digitization, BIM modeling, dynamic BIM creation, analysis engines, and object mapping to real-world hospitals.

5.2. DT models for diagnosis and treatment

Authors in [8] performed a detailed analysis of the current literature on the use of DTs in rhinology, including the advantages and limitations of this technology. In another study [76], a dietary and lifestyle program has been developed by Digbi Health to achieve weight loss that effectively reduces weight and improves diverse health outcomes by prioritizing and personalizing food ingredients to match an individual's genetic profile and nurturing the gut microbiome. In this study, they analyzed the weight loss pattern and microbiome profile of 103 individuals to identify the effects of the weight loss program on the gut microbiome between their baseline and follow-up samples. The study found that 80% of individuals lost weight during the study. A genotype-specific heart DT (Geno-DT) approach has been developed in [65] to investigate the role of pathophysiological remodeling in sustaining ventricular tachycardia (VT) reentrant circuits and to predict the VT circuits in Arrhythmogenic right ventricular cardiomyopathy (ARVC) patients of different genotypes. ARVC is a genetic cardiac disease that leads to VT, a life-threatening heart rhythm disorder. Treating ARVC remains challenging due to complex underlying arrhythmogenic mechanisms involving structural and electrophysiological (EP) remodeling. They integrated the patient's disease-induced structural remodeling reconstructed from contrast-enhanced magnetic-resonance imaging and genotype-specific cellular EP properties.

Authors in [77] proposed a DT technology that can exhaustively and methodologically analyze the associations between a physical cancer patient and a corresponding digital counterpart to isolate predictors of neurological sequelae of the disease. This proposition stemmed from the premise that data science can complement clinical acumen to scientifically inform the diagnostics, treatment planning, and prognostication of cancer care. Another study attempted to harness DT in detecting, forecasting, diagnosing, and treating Anaphylaxis shocks by simulating the broad range of organs in people at high risk for these shocks [78]. Authors in [34] performed a detailed study to summarize the progress made in the use of the DT for choosing the right drug for a person, as well as presenting a subnet model that predicts existing drugs for a particular pathology based on genetic expressions involved in the disease, and genes addressed by therapy.

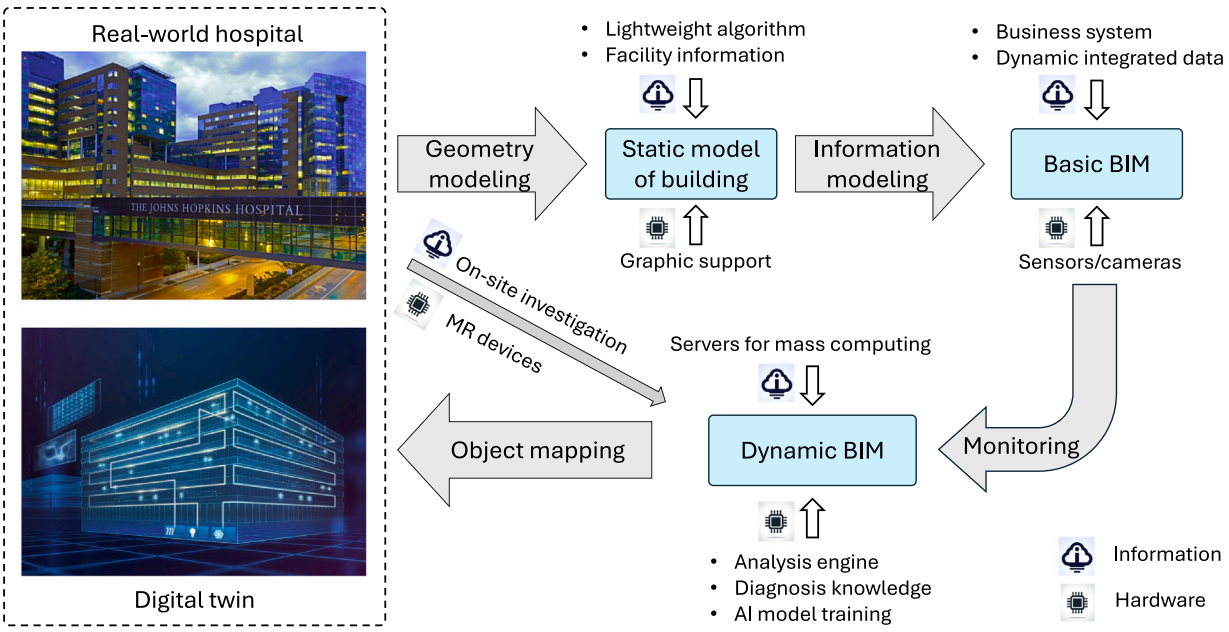


Fig. 7. Steps to establish a DT for a hospital building from civil engineering aspect [62].

5.3. DTs model for patient monitoring and management

In [59], two cloud-based and edge-based DT frameworks have been developed and compared in terms of scalability, flexibility, latency, and security. The authors extracted features over electrocardiogram signals and performed popular ML algorithms. They represented the frameworks by developing a case study to predict cardiac patients, continuously monitor the risks related to heart disease, and report the risk to both healthcare professionals and users in real-time. In another study [53], a system was developed for monitoring and preventing Cardiac pathologies using AI and IoT sensors, capable of facilitating the patient-doctor relationship, improving medical treatment times, and reducing costs. Authors in [61] developed a patient-specific dynamical system model from the time series data of the cancer patient's metabolic panel taken during cancer treatment and recovery. The model consisted of stacked LSTM recurrent neural networks and a fully connected neural network in each unit.

In [63], a methodology was developed that uses DES connected to the IoT devices to build a DT platform of patient pathways in a hospital for near real-time monitoring and predictive simulation. The authors validated their developed methodology using an experimental platform that mimics the behavior of a hospital. Authors in [55] investigated MRI-based DT models to forecast patient-specific treatment responses to Neoadjuvant Chemotherapy in triple-negative breast cancer. As reported, the developed model captured patient-specific, spatiotemporal dynamics of TNBC response to NAST, providing highly accurate predictions of NAST response. In another study [79], inspired by Leonardo DaVinci's Vitruvian Man, a DT of Patient platform was developed using Microservice Architecture (MSA), containerization (Docker), orchestration (Kubernetes), and ML Operations (MLOps). They designed the platform's architecture with multiple clusters of Docker containers and Kubernetes orchestration. Specific parts or organs of the human body were represented by clusters called DT-components (DTCs), and the set of those DTCs structure the "patient-DT" cluster in which appropriate pipelines define and monitor in real-time the "best" possible construction of the patient's DT.

5.4. DTs for personalized treatments

Authors in [80] developed a DT framework for precision neuromusculoskeletal health care built upon the International Standards Organization (ISO) framework. As reported, the developed framework presented remarkable computational models for clinical application. A review of the DT concept and available literature on DTs for health disorders such as neurological diseases have been investigated in [81]. They explored the potential and challenges of implementing intelligent DTs for the future management of migraines. In another study [82], the use of DT technology for remote motor rehabilitation, particularly focusing on robot-assisted exercises controlled through virtual reality, was explored. The authors proposed a concept of using a DT to operate a rehabilitation exoskeleton remotely, with safety systems predicting harmful situations. They found that this approach could be widely applied in telerehabilitation, offering potential benefits but also facing challenges in real-world implementation. Authors in [56] introduced a novel mathematical model as a DT to predict metabolic responses before and after long-term fasting, addressing the lack of consensus on diet impacts. The tool integrates data from various clinical studies and can predict new, independent data, including unmeasured variables like hepatic glycogen and gluconeogenesis. It adapts predictions based on an individual's sex, weight, height, and historical metabolic data.

In [35], an integrated informatics platform was developed to improve access to medical services for patients within smart communities. Using AI techniques, the authors introduced a model for representing each patient's health state, creating DT based on biological data gathered through sensor networks. These digital models supported decision-making for family doctors and medical experts. The platform combined data analysis, a priori patient information, and automated learning to enhance healthcare delivery. Authors in [83] proposed a MoSTHealth framework, utilizing DTs and Model-Driven Engineering (MDE) to address the growing demand for personalized infertility treatments. The framework allowed healthcare experts to model tailored Web of Things (WoT) healthcare monitoring systems (HMS) for individual patients, incorporating real-time data from devices monitoring hormones, heart rate, and temperature. The study presented a case scenario for infertility treatment, generating a service-oriented cloud architecture with ML-based prediction modules. In [58], a framework for integrating DTs into precision medicine was developed to enhance personalized disease management. The framework combined technologies such as AI, IoT, cloud computing, and cyber–physical systems to create a unified platform for analyzing vast amounts of health data. As reported, the developed framework allowed medical professionals to

conduct controlled trials and predict outcomes for different treatments. Authors in [57] presented a hybrid DT model for simulating ischemic stroke risk, combining mechanistic and ML approaches. The mechanistic model simulated the evolution of physiological risk factors, such as blood pressure and type 2 diabetes, under different interventions, such as diet and medication changes. They used their results by an ML model to calculate the 5-year stroke risk at various time points. The study highlighted the potential of DTs in patient education and clinical decision-making, promoting behavior change and improving adherence to treatment plans.

5.5. Developing DTs for medical devices

Authors in [84] developed a DT framework using AI to measure soft tissue deformation during head and neck surgery, addressing the challenge of soft tissue shift detection. Using 3D scans from HoloLens 2 and ArtecEva, soft tissue resection and the corresponding cavity were scanned in cold and warm conditions to simulate tissue shifts by temperature change. As reported, significant differences in volume measurements between the scanners were noted, but both provided reliable results. The study demonstrated that tissue shift could be measured, with the cavity showing greater sensitivity to temperature changes than the resected tissue. This approach has potential applications in surgical planning and tissue management. A DT for testing endoscopic medical devices was developed in [29] to reduce the need for time-consuming and resource-intensive animal studies and clinical trials. The DT was constructed to replicate the human gastrointestinal tract (GIT), enabling the study of interactions between devices and tissues. The authors highlighted the complex anatomy and processes of the GIT, emphasizing the potential of the DT to improve the design and evaluation of endoscopic devices, including robotic capsule endoscopes. Authors in [85] introduced the concept of data-driven DTs in laparoscopic surgery, utilizing augmented reality and ML to enhance minimally invasive procedures. The developed DT provided a virtual representation of patient and organ data, offering surgeons better access to complex information. Reportedly, their study highlighted challenges such as accurately mapping the DT to real-time conditions, addressing deformations and movements, and balancing intuitive user access with safety.

5.6. Other studies

Authors [86] introduced TWIN, a sample-efficient method for generating personalized DT of clinical trial patients. The DTs accurately reflected individual patient characteristics and preserved temporal relations across visits and events. According to the authors, TWIN generated high-fidelity clinical trial data even in low-data scenarios, unlike some methods that focused on Electronic Health Records (EHRs) and required large datasets. The study demonstrated that TWIN improved patient outcome predictions, reduced the need for extensive participant recruitment in trials, and offered strong privacy protection for actual patient data. According to another study /cite[chen2022digital], DT model (COVID-DT) can enhance epidemic prevention and control (P&C) based on AI algorithms, blockchains, and deep learning techniques. The model mapped real-world epidemic P&C efforts into a virtual space, utilizing BiLSTM for prediction accuracy and blockchain for data security. According to the authors, the developed COVID-DT model achieved stable data transmission rates, low latency, and high network security performance. Authors in [87] proposed a blockchain-based collaborative DT framework for decentralized epidemic alerting to address the challenges posed by the COVID-19 pandemic. The framework enabled secure real-time data exchange and analysis among multiple participants by integrating DTs, blockchain, and AI. In another study [88], a real-time emotion recognition (ER) system integrated with a DT model was developed for personalized healthcare. The system utilized ML and deep learning techniques to monitor and analyze a person's emotions, providing insights for improving health and well-being. The authors proposed an efficient framework using real-time image processing via a web camera to address challenges such as limited datasets and false emotion classification. As reported, the developed system demonstrated promising results with minimal training time and high accuracy, offering the potential for early disease diagnosis and effective treatment in healthcare centers.

5.7. Cross-study comparative analysis

Table 5 systematically compares the 34 representative DT implementations reviewed in this section across application domain, data sources, computational methods, and reported outcomes. The synthesis reveals several critical patterns:

- Validation Maturity Gradient: Only 6 of 34 studies (18%) report explicit quantitative accuracy metrics. The highest validated performance is observed in hybrid cloud–IoT architectures achieving 95%–99% accuracy on combined MIMIC-III and real-time sensor data [45], while edge-deployed cardiac DTs demonstrate 85.77% accuracy with a 4.8 s inference time [26]. The majority of implementations remain at the framework-proposal, simulation-only, or conceptual-study stage, without large-scale clinical validation.
- Dataset Scale Limitation: Reported cohort sizes range from highly individualized patient-specific models ($n < 20$) to moderate-scale datasets such as the PTB ECG database (n=200). No study reports validation on cohorts exceeding 1000 patients, fundamentally constraining claims of generalizability and hindering regulatory translation.
- Algorithmic Diversity: Mechanistic and physics-based modeling approaches dominate medical device evaluation and surgical planning applications, whereas data-driven ML/DL techniques prevail in patient monitoring and diagnostic tasks. Notably, only four studies adopt hybrid mechanistic–ML architectures [55–57, 65], despite their theoretical advantages in interpretability and physiological consistency.
- Deployment Readiness: Evidence of real-world clinical deployment remains limited. Among the 34 reviewed implementations, only five report operational use beyond experimental or pilot settings [26,44,45,62,76]. The remainder are confined to simulation environments, architectural frameworks, or single-site feasibility studies.
- Given the above investigations, the findings reinforce the translational gap discussed in Section 6 : although high technical performance (e.g., 95%–99% accuracy in controlled settings) is achievable, widespread clinical adoption is impeded by insufficient validation rigor, limited dataset scale, interoperability challenges, and unresolved regulatory considerations.

6. Discussions

The synthesis of findings across this review demonstrates that Digital Twins (DTs) are evolving from conceptual prototypes into computational infrastructures capable of transforming patient monitoring, diagnostic reasoning, and therapeutic decision-making. By coupling IoT-enabled data acquisition with adaptive AI models, DTs enable continuously updated digital representations that mirror physiological and operational states. This dynamic reciprocity between the physical and virtual domains represents a structural shift in healthcare from reactive, episodic care toward proactive, evidence-driven, and context-aware clinical intelligence [36,64,90].

Across implementations, the reviewed studies converge on three central observations. First, DTs consistently enhance predictive performance, diagnostic precision, and process optimization, providing a computational substrate for personalized and population-level healthcare analytics. Second, hybrid paradigms that integrate mechanistic

Table 5
Comparative performance analysis of healthcare DT implementations.

Study	Application domain	Dataset/Data source	Model/Algorithm	Validation method	Reported performance	Ref
Hospital Operations and Resource Management						
Hospital Intelligent Twins	Hospital operations	Facility sensors, BIM data	BIM + IoT analytics	Single-site deployment	Whole-facility situational awareness; no quantitative metrics	[62]
Hospital Resilience DT	Facility design	BIM + hazard models	Civil engineering simulation	Design validation	Qualitative resilience guidance; no performance metrics	[75]
Patient Pathway DT	Hospital flow	IoT devices + RTLS	DES + IoT integration	Experimental platform	Near real-time simulation; no quantitative validation	[63]
COVID-DT	Epidemic P&C	Epidemic signals	BiLSTM + blockchain	Simulation	Stable transmission, low latency; no specific metrics	[89]
Blockchain Epidemic Alert	Epidemic surveillance	Multi-party RT data	Blockchain + AI	Framework proposal	Secure data exchange; no deployment metrics	[87]
Diagnosis and Treatment Planning						
Cardio Twin (IHD)	Cardiac diagnosis	PTB database (n=200)	CNN (edge deployment)	Single-center evaluation	85.77% accuracy, 4.8 s per classification	[26]
ARVC VT Geno-DT	Cardiac arrhythmia	CE-MRI + genotype data	Genotype-specific EP modeling	Patient-specific	VT circuit prediction; no quantitative accuracy	[65]
Oncology Neuro-sequelae DT	Cancer complications	Multimodal oncology data	Data-science association analysis	Retrospective analysis	Predictor isolation; no predictive metrics	[77]
Drug Repurposing DT	Pharmacology	Gene expression databases	Subnet modeling	Computational validation	Framework demonstration; no clinical validation	[34]
Rhinology DT (review)	ENT applications	Literature review	Multiple approaches surveyed	Systematic review	Advantages/limitations synthesis; no new data	[8]
Patient Monitoring and Management						
Cardiac Risk DT (Edge vs Cloud)	Cardiac monitoring	ECG features (IoT)	Feature extraction + ML	Comparative framework	Real-time risk prediction; no accuracy reported	[59]
AI+IoT Cardiac Monitor	Cardiac pathology	IoT sensors	ML + wearable integration	System development	Improved treatment times; no quantitative metrics	[53]
Cancer Metabolic Panel DT	Oncology monitoring	Patient metabolic panels	Stacked LSTM + FC layers	Time-series modeling	Patient-specific dynamics captured; no validation metrics	[61]
TNBC Response DT	Cancer treatment	Longitudinal MRI (TNBC patients)	MRI spatiotemporal modeling	Patient-specific (n<50)	Highly accurate NACT response prediction; no specific %	[55]
Hybrid DT (Cloud-IoT)	Multi-parameter monitoring	MIMIC-III + IoT sensors	MLP + XGBoost	Hybrid architecture validation	95%–99% accuracy; 25% training latency reduction	[45]
Real-time DT Framework	Vital signs monitoring	SpO_2, HR, BT (IoT)	Pyomo optimization + Twin-graph	Multi-patient simulation	32% latency reduction, 52% response improvement	[44]
Context-Aware DT	Heart disease diagnosis	ECG rhythms	ML classifier + DT framework	Simulation study	Context-aware diagnosis; no accuracy reported	[22]
Personalized Treatments and Precision Medicine						
Neuromuscu-loskeletal DT	Precision rehab	Biomechanical + clinical data	ISO-aligned computational models	Framework proposal	Computational framework; no clinical validation	[80]
Migraine DT (review)	Neurological disease	Literature synthesis	DT + AI (conceptual)	Review study	Potential/challenges mapped; no implementation	[81]
Telerehab Exoskeleton DT	Motor rehabilitation	Kinematics + VR telemetry	Safety-aware VR control	Conceptual study	Remote exercises with hazard prediction; no trials	[82]
Metabolic Fasting DT	Dietary intervention	Clinical studies + demographics	Mathematical physiology model	Independent data prediction	Personalized response incl. unmeasured variables	[56]
Smart Community Health DT	Primary care	Wearables + environmental sensors	AI state models	Platform development	Decision support for family doctors; no validation	[35]
MoSTHealth Infertility DT	Reproductive health	Hormones, HR, temperature	Model-Driven Eng. + ML	Case scenario	Tailored monitoring + cloud architecture; no outcomes	[83]
Precision Medicine Framework	Multi-disease	Multimodal clinical data	AI/IoT/cloud/CPS fusion	Framework proposal	Controlled virtual trials; no real-world deployment	[58]
Hybrid Stroke Risk DT	Cardiovascular risk	Risk factors + interventions	Mechanistic + ML fusion	Simulation	5-year stroke risk prediction; no validation cohort	[57]
Weight Loss Microbiome DT	Metabolic health	Genotype, gut microbiome	Personalized program analysis	Longitudinal (n=103)	80% weight loss; microbiome shifts identified	[76]
Medical Devices and Surgical Applications						
Soft-Tissue Shift DT	H&N surgery	HoloLens2/ArtecEva scans	3D scanning + AI measurement	Cadaver/simula-tion	Temperature-dependent tissue shift quantified; no clinical use	[84]
Laparoscopic Surgery DT	Minimally invasive surgery	Patient/organ visual data	AR registration + ML	Conceptual framework	Enhanced situational awareness; mapping challenges noted	[85]

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Table 5 (*continued*).

Study	Application domain	Dataset/Data source	Model/Algorithm	Validation method	Reported performance	Ref
Endoscopic Device Testing DT	GI tract devices	Device–tissue interaction data	Physics-based GI simulation	Virtual testing	Reduced animal testing; benchmarking capability	[29]
Other Applications						
TWIN Clinical Trials	Trial simulation	Trial EHR time-series	Sample-efficient generative DT	Synthetic patient generation	High-fidelity synthetic patients; improved outcome prediction	[86]
Emotion Recognition DT	Mental health	Webcam video streams	CV/DL emotion recognition	Real-time system	Fast training, high accuracy; no specific %	[88]
Patient-DT Platform (Vitruvian)	Multi-organ modeling	Multimodal organ components	MSA/K8s/MLOps orchestration	Platform architecture	Real-time patient-specific DT composition; no clinical data	[79]
Federated DT (AoI-aware)	Multi-department collaboration	Departmental clinical data	FL + split learning + DRL	Simulation study	50% AoI reduction; privacy-preserving	[47]

modeling with machine learning show superior interpretability and generalization compared to purely data-driven approaches, reinforcing the role of physics-informed learning in clinical trustworthiness. Third, the emergence of federated, edge, and privacy-preserving infrastructures signals a methodological transition from centralized architectures to distributed, autonomy-preserving intelligence aligned with modern healthcare governance models.

In Section 5, recent works on DTs in healthcare were methodologically classified by considering several case studies under different categories, such as DTs for hospitals, diagnosis and treatment, patient monitoring and management, personalized therapies, and medical devices. This section delves deeper into specific applications and implementations of DT technologies in healthcare, offering a comprehensive review of case studies across multiple domains. Through systematic categorization, it highlights the progress achieved in diverse healthcare contexts where DTs have been deployed, demonstrating their transformative potential across operational, clinical, and therapeutic workflows.

Beyond conventional domain-based groupings, we classify applications by their analytic task, which better reflects the objectives and algorithms that drive each twin. This perspective surfaces shared methods across domains and supports a consistent basis for comparison. The review is structured around four task families: simulation/optimization, classification/diagnosis, prediction/prognosis, and control/decision support.

- Simulation and Optimization: Virtual replication of physiological systems or hospital operations for experimentation, design validation, or workflow optimization (e.g., hospital intelligent twins, surgical rehearsal systems).
- Classification and Diagnosis: Data-driven identification of disease states or patient subtypes using supervised learning and pattern recognition (e.g., CNN-based cardio twins, cancer subtype modeling).
- Prediction and Prognosis: Forecasting disease progression, treatment response, or system behavior via temporal or probabilistic models (e.g., LSTM- or RL-enabled cardiac monitoring, metabolic forecasting).
- Control and Decision Support: Closed-loop or semi-autonomous intervention through adaptive controllers, reinforcement learning, or edge-intelligent systems (e.g., rehabilitation robots, infusion and energy control twins).

This conceptual framework serves two purposes: (i) it enables cross-domain synthesis by aligning application areas with computational goals, and (ii) it supports a more systematic evaluation of the underlying AI/ML methodologies that power healthcare DTs. Accordingly, Table 6 summarizes representative use cases across these analytic dimensions linking each to its domain, core technique, observed outcome, and reference, to facilitate a unified interpretation of DT-driven innovations in healthcare.

Recent experimental evaluations support these conceptual insights with measurable performance gains. Jameil et al. (2024) implemented a Pyomo-based optimization framework for real-time DT orchestration and reported a 32% reduction in telemetry latency and a 52% improvement in system response time under high-load scenarios [44]. Similarly, Jameil et al. (2025) demonstrated that integrating Multilayer Perceptron (MLP) and XGBoost models within a hybrid cloud–IoT DT architecture reduced model-training latency by 25% and improved testing efficiency by 33%, while maintaining 95%–99% predictive accuracy across MIMIC-III and IoT datasets [45]. These quantitative results substantiate that modern DT frameworks are transitioning from simulation-based experimentation to operationally validated, latency-constrained deployments. According to a systematic meta-analysis of 192 different limitations extracted from 11 recent studies, over 500 references (2022–2025), fundamental barriers persist despite these technical advances. The analysis identified nine primary limitation categories, with data integration and interoperability emerging as the universal bottleneck, appearing in 100% of reviewed papers with the maximum impact severity score. This finding reinforces those technical capabilities in controlled settings (achieving 95%–99% accuracy) exist alongside critical deployment barriers that prevent translation to clinical practice [91–93].

From a system perspective, DT deployments now span a continuum, ranging from hospital-scale operational twins that support resource management to patient-specific diagnostic and therapeutic twins that drive individualized interventions. This heterogeneity underscores the versatility of DT frameworks but also reveals persistent fragmentation in architectural design, data semantics, and performance reporting. Inconsistent standards, disparate interoperability protocols, and heterogeneous validation procedures impede reproducibility and regulatory scalability across healthcare systems. Notably, only about 32% of identified data integration challenges included proposed future directions, indicating a fundamental lack of consensus on solutions and suggesting these barriers require breakthrough innovations rather than incremental improvements [94,95]. Hence, advancing DTs from research prototypes to safety-critical infrastructure demands a harmonized interplay between technical, ethical, and organizational dimensions.

To systematically assess deployment challenges, we conducted a quantitative meta-analysis of limitations across the reviewed literature (Fig. 8). The comparative challenge matrix reveals a critical paradox: challenges receiving the highest research attention often have the lowest solution availability. Data integration, despite substantial research focus, shows predominantly “Low” to “Medium-Low” solution availability, indicating the need for breakthrough innovations rather than incremental improvements. This creates an inverted priority structure where foundational technical challenges (Data Integration, Model Validation, Computational Complexity) remain largely unresolved despite high attention, while regulatory and clinical validation shows more mature solutions. This disparity suggests a clearer consensus on regulatory pathways than on fundamental technical problems. In the next section, we discuss the most important limitations.

Table 6
Representative digital twins use cases in healthcare organized by analytic task.

Application domain	Analytic task (Framework)	AI/ML/Core technique	Primary inputs/data	Observed outcome (as reported)	Ref.
Smart hospital ops (Hospital Intelligent Twins)	Simulation/Optimization	BIM, IoT analytics, dashboarding	Facility sensors, dept. KPIs	Whole-hospital situational awareness; resource/energy management; ward/device interactivity	[62]
Hospital resilience/facilities	Simulation/Optimization	DT for civil/infra modeling	BIM + hazard models	Design guidance for extreme-weather resilience; continuity of care impact framing	[75]
Rhinology (review scope)	Classification/Prediction (surveyed)	Imaging + DT pipelines (lit. review)	ENT imaging/clinical data	Advantages/limits synthesis; implementation gaps identified	[8]
Genotype-specific cardiac DT (ARVC VT)	Prediction/Classification	Genotype-specific modeling, EP simul.	CE-MRI remodeling + cellular EP	VT circuit prediction across genotypes; mechanistic insight into reentry	[65]
Weight-loss program (microbiome)	Prediction/Classification	Microbiome + personalized program	Genotype, gut microbiome, outcomes	80% participants lost weight; microbiome shifts associated with program	[76]
Oncology neuro-sequelae DT	Prediction/Prognosis	Data-science twin association analysis	Multimodal oncology data	Predictors of neurological sequelae isolated for planning/prognosis	[77]
Anaphylaxis risk DT	Prediction/Control	Multiorgan simulation	Physiological signals/risk factors	Early detect/forecast; therapy simulation for high-risk individuals	[78]
Drug repurposing/selection	Prediction/Classification	Subnet DT over gene expression	Gene–disease–drug graphs	Pathology-specific drug prediction framework	[34]
Cardiac risk DT (cloud vs edge)	Prediction/Control	Feature extraction + classic ML	ECG features (IoT)	Real-time risk prediction; latency/security tradeoff (edge wins for delay)	[59]
AI+IoT cardiac monitoring	Prediction/Control	ML over wearable/IoT streams	ECG + vitals	Faster medical response; improved treatment times/costs	[53]
Metabolic panel time-series DT (cancer)	Prediction/Prognosis	Stacked LSTM + FC layers	Longitudinal lab panels	Patient-specific dynamics captured during therapy/recovery	[61]
Patient pathway DT (hospital flow)	Simulation/Prediction	DES + IoT integration	RTLS/IoT pathway events	Near real-time flow simulation for operational prediction	[63]
MRI-driven TNBC NACT response	Prediction/Prognosis	MRI-based spatiotemporal DT	Longitudinal MRI (TNBC)	Accurate patient-specific response forecasting to NACT	[55]
Patient-DT platform (MSA/K8s/MLOps)	Control/Decision Support	Microservices, orchestration pipelines	Multiorgan DT components (DTCs)	Real-time composition/monitoring of patient-specific DT	[79]
Neuromusculoskeletal precision care	Simulation/Optimization	ISO-aligned computational models	Biomech. + clinical data	Framework for precision rehab/orthopedics	[80]
Intelligent DTs for migraine (review)	Prediction/Control (surveyed)	DT + AI (conceptual)	Patient + trigger data	Potential/future challenges mapped	[81]
Telerehab exoskeleton DT (VR)	Control/Decision Support	Safety-aware control via DT	Kinematics + VR telemetry	Remote robot-assisted exercises; proactive hazard prediction	[82]
Metabolic fasting DT	Prediction/Simulation	Mathematical physiology model	Clinical studies + demographics	Personalized fasting response incl. unmeasured vars (hepatic glycogen)	[56]
Smart-community health DT	Decision Support/Prediction	AI state models over sensors	Wearables + env. sensors	Primary care decision support in communities	[35]
MoSTHealth (infertility)	Simulation/Prediction	Model-Driven Eng. + ML modules	Hormones, HR, temperature	Tailored infertility monitoring + service-oriented cloud architecture	[83]
Precision medicine integration DT	Simulation/Prediction	AI/IoT/cloud/CPS fusion	Multimodal clinical data	Controlled virtual trials; outcome prediction	[58]
Hybrid stroke-risk DT	Prediction/Decision Support	Mechanistic + ML fusion	Risk factors + interventions	5-yr stroke risk under scenarios; patient education utility	[57]
Soft-tissue shift (H&N surgery)	Simulation/Control	3D scans + AI measurement	HoloLens2/ArtecEva scans	Quantified temperature-dependent tissue/cavity shift for planning	[84]
Endoscopic device testing (GI DT)	Simulation/Optimization	Physics-based GI tract twin	Device–tissue interaction	Benchmarked endoscopes; reduced animal/clinical testing burden	[29]
Laparoscopic surgery DT (AR+ML)	Control/Decision Support	AR registration + ML	Patient/organ visual data	Enhanced intraop situational awareness; mapping/ deformation challenges	[85]

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Table 6 (*continued*).

Application domain	Analytic task (Framework)	AI/ML/Core technique	Primary inputs/data	Observed outcome (as reported)	Ref.
TWIN: clinical trial twins	Prediction/Simulation	Sample-efficient generative DT	Trial EHR time-series	High-fidelity synthetic patients; better outcome prediction with low data	[86]
COVID-DT (epidemic P&C)	Prediction/Simulation	BiLSTM + blockchain	Epidemic signals + logs	Stable rates, low latency; secure P&C coordination	[89]
Decentralized epidemic alerting	Prediction/Simulation	Blockchain + DT + AI	Multi-party RT data	Secure real-time exchange/analysis across participants	[87]
Emotion-aware DT for care	Classification/Decision Support	CV/DL emotion recognition	Webcam video streams	Fast training; high accuracy; early dx potential for centers	[88]

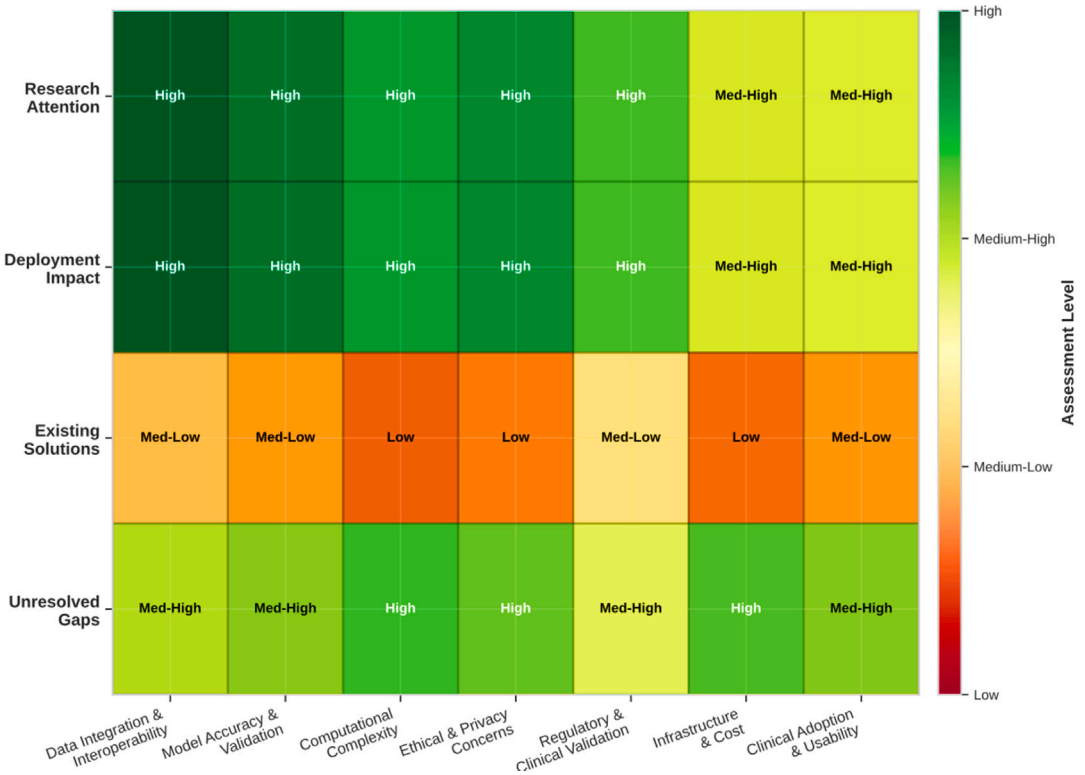


Fig. 8. Comparative Challenge Matrix—multi-dimensional assessment of the top challenges. Columns list the top categories (sorted by normalized impact), and rows encode four dimensions: *Research Attention* (frequency in the literature), *Deployment Impact* (anticipated real-world significance), *Existing Solutions* (tools/methods available), and *Unresolved Gaps* (remaining challenges). Cell codes L/M−/M+ /H map to Low/Medium−/Medium+ /High on a 0–3 scale.

6.1. Limitations

Healthcare DTs promise transformative clinical capabilities, yet their translation from research prototypes to operational clinical systems remains constrained by a constellation of interconnected technical, regulatory, and organizational barriers. Systematic analysis of limitations reported across the surveyed literature reveals quantifiable patterns in both the frequency and severity of these challenges, enabling evidence-based prioritization of research directions. This subsection examines five critical limitation domains — data integration and interoperability, model accuracy and validation, privacy and ethics, computational complexity and scalability, and regulatory alignment — synthesizing 192 distinct limitations identified across 11 papers to characterize the current state of the field and illuminate pathways toward clinical translation.

6.1.1. The foundational challenge: Data integration and interoperability

Four interlocking barriers consistently emerge as determinants of successful DT translation. First, *interoperability* remains foundational: heterogeneous EHR systems, medical imaging archives, wearable sensors, and bedside monitors produce data with incompatible schemas and semantics, undermining real-time synchronization and multi-site portability [64,91,95]. The severity of this limitation is quantitatively confirmed, appearing 37 times across all analyzed papers, with researchers expressing high confidence (2.65/3.0) that it represents the field's most critical challenge. The universal recognition (100% paper coverage) combined with maximum impact score (100/100) indicates that without addressing data integration, advances in model accuracy, privacy preservation, and computational efficiency cannot be fully realized in clinical settings [94,96].

Architectural consensus increasingly favors layered, standards-first frameworks with explicit interfaces for data ingestion, semantic mediation, model analytics, and assurance layers aligned with healthcare-native standards such as HL7-FHIR and DICOM [68,90,91]. Without such harmonization, multimodal fusion remains brittle and error-prone in high-acuity pathways [94]. Jameil et al. (2025) further confirmed that delayed interoperability between edge and cloud nodes can degrade predictive accuracy by over 20%, reinforcing the need for FHIR-compliant data mediation and dynamic API orchestration within

healthcare DT infrastructures [45]. This finding gains additional significance when contextualized against the broader literature: researchers collectively identified that fragmented healthcare systems with heterogeneous data formats, inability to integrate wearable sensors with EHR systems, and missing standards for continuous real-time data synchronization constitute the primary obstacles to DT scalability [91,94,95].

The complexity extends beyond technical integration to semantic interoperability. The analyzed literature shows that handling multimodal medical data, including diagnostic imaging, physiological measurements, genomics, and longitudinal clinical records, poses fundamental challenges that current standards only partially address. This heterogeneity in data modalities compounds the difficulty of maintaining synchronized, coherent digital representations across distributed healthcare environments [36,64,90].

6.1.2. Model accuracy, validation, and the translation gap

The second-most critical barrier, model accuracy and validation, was identified 33 times across all papers with an impact score of 78.6/100. This reveals a fundamental paradox: while DT systems demonstrate impressive accuracy in controlled experimental settings (95%–99% as reported above), their real-world clinical validation remains severely limited. Of the 33 validation challenges identified, researchers expressed medium-to-high confidence (2.33/3.0) that this gap is genuine and significant, yet only 27.3% of these limitations included proposed future research directions [92,93,95].

This validation paradox manifests in multiple dimensions. First, current DT models exhibit dependence on assumptions that inadequately reflect real-world clinical conditions, with mathematical modeling approaches particularly vulnerable to simplifications that compromise accuracy in heterogeneous patient populations [36,91]. Second, experimental modeling approaches suffer from calibration errors that propagate through prediction pipelines, especially in noisy or non-IID data environments where performance can degrade by 20% or more [45,47]. Third, the literature reveals limited validation in diverse clinical scenarios, most implementations are evaluated on fewer than 20 participants or synthetic datasets, fundamentally constraining generalizability claims [92,93].

Furthermore, many current implementations oversimplify complex physiological processes to achieve computational tractability, potentially missing critical interactions that determine clinical outcomes. This reductionism represents a methodological tension between computational feasibility and clinical fidelity that has not been adequately resolved in the literature [90,91]. The hybrid mechanistic-ML approaches noted earlier partially address this concern by anchoring data-driven learning within physiologically plausible constraints, yet comprehensive validation frameworks that systematically evaluate these hybrid models across diverse patient populations remain absent [36,64].

6.1.3. Privacy, ethics, and the security-performance tradeoff

Privacy and ethics define the societal and regulatory boundaries of DT scalability. Federated learning (FL) enables localized training without central aggregation, mitigating privacy risks but introducing fairness and convergence challenges under non-IID conditions [47,95]. Distributed-ledger approaches strengthen provenance and auditability but increase latency and energy cost, often conflicting with the responsiveness requirements of clinical decision support [96]. Ethical deployment therefore requires explicit data-use transparency, consent traceability, and explainable AI pipelines to sustain public and professional trust [92,93]. The multi-layer threat model proposed by Jameil et al. (2025) demonstrated that combining RSA encryption, TLS protocols, and role-based access control (RBAC) enhances security robustness but increases computational overhead, highlighting the need for lightweight cryptography and context-aware security management within real-time DT pipelines [45].

The systematic analysis reveals that ethical and privacy concerns were mentioned 30 times across all papers, earning an impact score of 77.6/100, nearly equivalent to model validation in severity. This indicates that privacy is not an implementation afterthought but a core technical and societal challenge preventing clinical deployment. The literature consistently identifies sensitive patient data sharing across departments and institutions as creating substantial privacy concerns during model training, particularly in federated settings where data ownership and consent boundaries span multiple jurisdictions [47,95,96].

Critical obstacles extend beyond encryption mechanisms to include compliance challenges with HIPAA, GDPR, and emerging AI-specific regulations; trust deficits in federated learning frameworks where participants cannot verify remote data quality or usage; and persistent risks of unauthorized access to vulnerable health information despite multilayer security architectures [92,93,96]. Notably, only 23.3% of identified privacy limitations included proposed future directions, suggesting the field lacks mature frameworks for balancing privacy preservation with clinical utility—a tension that fundamentally constrains scalability of distributed DT infrastructures. This privacy–utility tradeoff creates a false dichotomy that the field has not yet resolved. Privacy-preserving techniques such as differential privacy, homomorphic encryption, and secure multi-party computation provide strong mathematical guarantees, but they also introduce computational complexity that degrades real-time performance. Conversely, reducing privacy protections to achieve responsiveness violates regulatory requirements and erodes patient trust. Current implementations oscillate between these extremes without systematic frameworks for context-aware privacy calibration that adapts protections to data sensitivity, clinical urgency, and regulatory context [47,96].

6.1.4. Computational complexity, scalability, and infrastructure constraints

Computational scalability and latency represent tangible bottlenecks. Continuous synchronization of high frequency biosignals and imaging workloads imposes heavy computational burdens on conventional hospital IT infrastructures [64,90]. Hybrid edge–cloud architectures with Age-of-Information (AoI)–aware scheduling reduce information staleness and align computation with clinical urgency [47]. Yet such distributed deployments require disciplined synchronization, data compression, and priority-aware refresh policies. Security or provenance mechanisms (e.g., blockchain) must adopt energy-efficient consensus protocols to prevent responsiveness degradation in time-critical workflows [96]. Experimental results in [44] verify that dynamic resource scheduling and multi-threaded optimization can sustain sub-200 ms end-to-end response times even under high telemetry density, confirming the feasibility of AoI-driven orchestration for clinical DTs.

Computational complexity and scalability limitations were identified 31 times across 91% of analyzed papers (10 of 11), achieving an impact score of 72.4/100. This near-universal recognition demonstrates that even when data integration and validation challenges are resolved, sheer computational demands pose distinct barriers to clinical deployment. The literature reveals several persistent bottlenecks: inability to meet response speed requirements in emergency scenarios where seconds determine outcomes; high latency in blockchain-based security systems that delay critical decisions; insufficient edge computing resources for real-time processing of multimodal data streams; challenges managing diverse data distributions across federated healthcare systems; and difficulty handling multiple concurrent users with low-latency requirements [47,64,90,96].

The infrastructure dichotomy compounds these challenges: edge computing offers low latency critical for emergency care but suffers from limited storage capacity and computational power, while cloud computing provides scalability but introduces latency and security vulnerabilities. Infrastructure and cost considerations emerged 18 times across approximately 73% of papers, revealing that financial and physical constraints affect practical implementation even when technical

solutions exist [45,90,91]. Specific barriers include heavy reliance on cloud computing infrastructure requiring substantial investment; resource limitations in resource-constrained healthcare settings, particularly in low- and middle-income countries; high costs for implementation and maintenance that exceed many institutional budgets; dependence on specialized institutional support and facilities; and limited applicability in resource-poor settings where DT benefits could be most transformative [92,93].

Only 19.4% of computational complexity limitations included proposed solutions, reinforcing that current approaches lack systematic frameworks for balancing latency, energy efficiency, security, and cost. Most papers identify the problem but offer only architectural variations (edge vs. cloud) without quantitative optimization frameworks that explicitly trade off competing requirements based on clinical context, patient acuity, and available resources.

6.1.5. Regulatory alignment, clinical validation, and translation pathways

Regulatory alignment and clinical validation remain the most significant translational hurdles. Current evidence largely stems from simulation-based or single-center feasibility studies lacking long-term safety, effectiveness, and change-controlled verification. Regulatory agencies increasingly emphasize DT-specific evaluation frameworks linking model risk class to staged evidence generation from in silico validation and shadow mode testing to real-world performance monitoring [91,92]. Without such structured evaluation pipelines, uncertainty in approval pathways continues to constrain clinical deployment and investment [93,95]. Both Jameil et al. (2024, 2025) explicitly note the absence of multi-site trials and large-scale user validation in current DT studies, emphasizing that most implementations are evaluated on fewer than 20 participants or synthetic datasets, thus reinforcing our conclusion on the need for standardized clinical validation pipelines.

Regulatory and clinical validation challenges were reported 22 times across all papers with an impact score of 54.1/100. While appearing in every study, it received lower frequency than data integration or model validation, suggesting researchers view it as solvable once fundamental technical issues are addressed. However, this perspective may underestimate the complexity of regulatory translation [92,93,95]. The literature reveals several critical gaps: absence of standardized reference designs for healthcare DTs that regulatory bodies can evaluate consistently; limited clinical trial implementations with most systems remaining in laboratory or pilot stages; insufficient regulatory frameworks specific to DT technology, forcing application of device or software regulations that inadequately address DT-specific risks; challenges ensuring clinical reliability and diversity of synthetic or augmented training data; and need for authorized institutional oversight and approval processes that balance innovation with patient safety [90–92].

Notably, regulatory validation showed the highest solution proposal rate at 40.9% (9 solutions for 22 challenges), suggesting clearer, more actionable pathways exist compared to foundational technical barriers. Proposed directions include establishing standardized validation frameworks adapted from medical device regulations; developing multi-phase implementation pipelines progressing from in silico validation through shadow deployment to active clinical integration; enhancing interdisciplinary collaboration between clinicians, engineers, ethicists, and regulators; and engaging regulatory bodies early in DT development to co-create evaluation frameworks [91,92,95].

Clinical adoption and usability concerns, mentioned 19 times across 82% of papers, compound regulatory challenges. Limited practical demonstrations of patient outcome improvements, fragmented healthcare delivery systems, insufficient user-friendly interfaces for clinicians, digital skill gaps among healthcare providers, and resistance to workflow changes collectively create organizational barriers that technical excellence alone cannot overcome [92,93,95]. These findings underscore that successful DT translation requires not only technical validation but sociotechnical implementation science addressing change management, training, workflow integration, and organizational readiness.

6.2. Future works

Taken together, these findings underscore that realizing clinically reliable DTs requires a multi-dimensional synthesis of interoperability, privacy, scalability, and regulatory assurance. Standards-first architectures, federated privacy-preserving analytics, and edge-intelligent operations, coupled with quantitative interoperability metrics, latency budgets, and post-market surveillance, represent the foundation of trustworthy DT ecosystems.

The systematic limitations analysis reveals a hierarchical dependency structure among these challenges: data integration represents a foundational barrier (impact score 100/100) that must be resolved before higher-level concerns like clinical adoption (impact score lower) can be fully addressed. This dependency suggests that targeted investment in data standardization protocols, particularly FHIR-compliant semantic mediation, real-time synchronization frameworks, and multimodal fusion architectures, could unlock progress across multiple limitation categories simultaneously [90,91,95].

Furthermore, the low solution proposal rates for foundational challenges (27%–32%) versus regulatory concerns (41%) indicate an inverted priority structure in current research: the field demonstrates clearer consensus on regulatory pathways than on solving fundamental technical problems. This pattern suggests that advancing DT maturity requires rebalancing research investment toward foundational interoperability and validation science, potentially through coordinated multi-institutional initiatives that develop and validate common architectures, shared datasets, and standardized evaluation protocols [36,64,92].

The field must also address the temporal dimension of validation: current studies report performance at single timepoints or over short evaluation windows, but clinical DTs must maintain accuracy, safety, and reliability over months to years as patient physiology evolves, treatment regimens change, and system components are updated. This longitudinal validation requirement, addressing model drift, calibration decay, and continuous learning under distribution shift, remains largely unaddressed in the literature [91,93]. Aligning these dimensions through shared governance, transparent validation, and explicit quantification of latency–privacy–energy–cost tradeoffs can accelerate the transition of DTs from experimental pilots to dependable, patient-centered infrastructure for precision healthcare. The convergence of high technical capability (95%–99% accuracy in controlled settings) with persistent deployment barriers (100% of papers identifying critical limitations) indicates the field has reached an inflection point: further technical refinement alone will not drive clinical translation. Instead, coordinated efforts addressing interoperability standardization, regulatory co-creation, sociotechnical implementation science, and long-term validation frameworks are required to realize DT's transformative potential in healthcare delivery.

7. Conclusion

The convergence of DT technology with IoT and AI is reshaping the foundations of healthcare delivery, shifting the field from episodic, reactive interventions toward more continuous, proactive, and personalized models of care. This review has synthesized the current state of DT research across clinical and operational domains, illustrating how DTs have progressed from conceptual constructs to functional systems capable of supporting real-time clinical decision-making, patient monitoring, and workflow optimization.

Across the literature, DT implementations demonstrate measurable technical gains, including high predictive accuracy, reduced telemetry latency, and improved system responsiveness, enabled by advances in multimodal sensing, federated and privacy-preserving learning, and hybrid mechanistic–machine learning modeling. These developments highlight the growing feasibility of real-time, data-driven clinical intelligence and underscore the significant potential of DTs to enhance both patient outcomes and healthcare system efficiency.

Yet the findings of this review also reveal a consistent set of translational barriers that constrain large-scale clinical adoption. Interoperability emerged as the most pervasive and severe challenge, appearing across all examined studies and influencing every stage of the DT lifecycle, from data acquisition to model validation and regulatory approval. Fragmented data architectures, limited semantic alignment across systems and the absence of standardized real-time exchange protocols collectively restrict the deployment of adaptive DT solutions in routine care. Related challenges, including privacy and governance concerns, computational demands, and the lack of validated evaluation pathways, further highlight the need for coordinated progress across technical and organizational domains.

Moving forward, meaningful advancement will require unified data standards, secure-by-design architectures, and scalable hybrid cloud–edge infrastructures that support continuous model updating while maintaining regulatory compliance. Equally important are rigorous validation frameworks, common benchmarking resources, and human-centered design approaches that promote clinician trust and seamless workflow integration. Our analysis indicates that foundational issues in data integration and model verification receive fewer concrete solution proposals compared to regulatory or organizational challenges, suggesting the need to rebalance research efforts toward these core technical bottlenecks.

In conclusion, the field stands at a pivotal moment. DT technology has demonstrated clear technical capability, but its translation into everyday clinical practice depends on coordinated action across research institutions, healthcare organizations, and regulatory bodies. With sustained investment in interoperability, validation science, and human-centered implementation, DTs have the potential to become key components of next-generation healthcare systems, enabling more predictive, personalized, and efficient care delivery at scale.

CRediT authorship contribution statement

Sara Shahnazinia: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Conceptualization. **Mahsa Tavasoli:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology. **Abdolhossien Sarrafzadeh:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Ali Karimoddini:** Writing – review & editing, Validation, Supervision, Funding acquisition.

Declaration of Generative AI and AI-assisted technologies in the writing process

Large language models were used solely to assist with grammar editing and clarity improvements. All technical content, analyses, and conclusions were independently developed, verified, and validated by the authors without AI assistance. The authors take full responsibility for the integrity and accuracy of the final manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to express their appreciation of the support from the North Carolina A&T State University Title III for sponsoring this research.

Data availability

No data was used for the research described in the article.

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